

Master's Programme in Building Technology

A Harmonized Classification System for Construction Products and Elements: The Designers' Perspective

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Abstract

This thesis investigates how construction products and elements are classified and proposes a prototype of a harmonized, designer-oriented classification system. The literature review explains the role of classification systems and builds the foundational understanding for the prototype to be harmonized, aligned with existing standards and designer needs by synthesizing the research. Additionally, thematic and comparative analysis of widely used classification systems shows that no single system simultaneously provides broad scope, fine granularity, and straightforward data exchange while also supporting everyday designer workflows. To address the identified gap, an action research approach is used with three iterative cycles, in which the classification system prototype is developed collaboratively with practitioners representing fields of structural, architectural, and MEP design, as well as BIM coordination. Additional insight is gathered through interviews, where input is acquired in regard to classification system usage and designer practices such as product discovery.

The outcome is an illustrative prototype classification system that implements classification through additive tag sequences from three parallel hierarchies (Element, Function, and Material) and enables multi-perspective navigation. The schema combines the taxonomy with discipline tags, machine and human-readable code, linking to attributes such as performance characteristics and other, flexible mapping to external classification systems. The prototype is intended to enable better product discovery, support other designer tasks, and facilitate structured data exchange while augmenting established classification systems. Practitioners judged the concept intuitive and useful for typical designer tasks. The main limitations include the illustrative scope of the current taxonomy and a modest practitioner sample. Further work should focus on broadening the coverage, continue attribute set formalization using normative sources, extend classification mappings, create an alias table for international translations, and validate the performance through pilot UI and BIM integrations.

Keywords classification system, BIM, interoperability, harmonization, product discovery, ISO 81346

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Abbreviations

API	Application Programming Interface
AR	Action Research
BIM	Building Information Modeling
BOQ	Bill of Quantities
bSDD	buildingSMART Data Dictionary
CAD	Computer Aided Design
CDE	Common Data Environment
COBie	Construction Operations Building Information Exchange
DoP	Declaration of Performance
EPD	Environmental Product Declaration
GWP	Global Warming Potential
HVAC	Heating, Ventilation, and Air Conditioning
IFC	Industry Foundation Classes
LCA	Life Cycle Assessment
MEP	Mechanical, Electrical, and Plumbing
UI	User Interface
WBS	Work Breakdown Structure

1 Introduction

1.1 Background

With the development of the construction industry Building Information Modeling (BIM) has become widely adopted. The benefits of BIM over conventional Computer Aided Design (CAD), which has been an essential part of any construction project for a long time, have been extensively discussed and acknowledged. In many developed countries legislation and institutional frameworks already support BIM, while as noted by Chatsuwan *et al.* (2024), developing countries, especially those with strong institutional support, also have standards in use, that are aligned with BIM practices. With these advancements, the focus is now shifting towards increasing the efficiency of BIM design workflows.

One of the key issues in this context is the integration of data for the sake of more flexibility during design and other benefits, as well as information flow to procurement and construction processes, which is being brought up directly or indirectly in current discussions within the industry. As mentioned by Alhava *et al.* (2024, p. 6), the basis for automated information flow is in harmonization of the available information itself throughout the construction phases. This challenge has a strong connection to classification systems used in construction, as they are the common ground in information flow.

However, existing classification systems often have shortcomings or problems regarding some applications. According to Friborg *et al.* (2017, p. 8) existing classification systems often suffer from unclear definitions, enumerative approaches, where concept, terms and synonyms mix in the same table, making it difficult to find or place objects. Viewing the challenges from the perspective of designers, the focus is set on products and elements as the main objects of design.

The term “product” refers to a specific item, material, or system manufactured by a supplier, whereas “element” refers to a generic building part or system, such as a wall, floor or beam. In this study, the combined term “products and elements” is used to cover both specific manufactured items and larger systems that they may be a part of.

Solving the challenges requires a framework of classifying products and elements for their management and maintenance. Meanwhile, the construction processes, which span from initial planning to the end of a structure's life cycle, can benefit from tracking construction products and

elements through all of the stages. The challenges mentioned are closely linked and must be approached with adaptability in mind, as a framework used for design purposes can lead to streamlining of procurement and other stages.

1.2 Objectives and research questions

The goal of this research is to build understanding and a foundation for a harmonized and user-friendly classification system for construction products and elements, while also outlining the process of the development and provide reasoning for choices made. The classification system is aimed to be developed from the perspective of designers.

The research is carried out to also help with the goal of ProdLib to streamline designers' workflows by improving the consistency and usability of its own system by developing a harmonized classification system. The system can serve as a universal tool for creating an interface between various services. A well-organized classification system, that takes inspiration from already used systems in the industry for these products and elements offers many advantages, including search for similar products and enhancing interoperability between different construction tools and databases.

In this thesis, the practical value of a classification system is assessed largely in terms of how well it supports common tasks facing a designer. Such tasks include product discovery, documentation, and decision making. These tasks are also linked to interoperability needs, for which classifications persist across different tools and project stages.

The outcome is expected to simplify the organization of materials and products within digital libraries, facilitate connectivity between product catalogues of various services, and enhance integration with BIM libraries, such as ProdLib, as well as other design software, quantity take-off tools, and LCA analysis solutions. One of the key aims is to ensure that the classification system is not only detailed and compatible with existing national systems but also intuitive and easy to use.

Sub-objectives of the thesis include:

1. Understand current classification system implementations, the problems they solve and their technical details.
2. Establish ways of creating compatibility with existing national and international systems.
3. Establish relevant data to enrich the system and support better organization and usability in design workflows.

4. Develop a harmonized classification system that simplifies the organization of construction materials and products from the perspective of designers.
5. Outline the process of creating a classification system, with emphasis on the steps needed for harmonization and feedback incorporation.

In order to achieve the objectives, six research questions were formulated:

1. What are the current classifications of construction products and elements used in the European Union and other key regions?
2. How can there be established a connection between unique classification systems that are utilized by specific regions or institutions?
3. What are the preferences of design offices and other users regarding material and product classifications?
4. How can item classification be linked to characteristic product and element attributes, and what benefits can this provide for designers and the construction process?
5. Can a classification system support both structured data exchange and flexible product discovery across different project stages?
6. How can feedback from practitioners be used to incrementally improve a classification system and enhance its usability in design workflows?

The research questions are structured to progress from understanding the existing practices related to classification systems, through focusing on practical user needs, to addressing technical challenges, and finally to refining and evaluating improvements through feedback. The first two questions build the understanding of how construction products and elements are currently being classified in Europe and other key regions, in addition to how national and institutional systems, which are different in scope and other aspects, might be linked together. The third research question focuses on the practical side, by inquiring preferences of design offices and other users in regards to classification, due to their adoption being most relevant for classification system success. Questions four and five help with solving technical requirements, such as the link between classifications and product attributes, and the ability to support both structured data exchange and flexible product discovery. Question six reflects the action research approach by inquiring how practitioner feedback can be used to iteratively refine and evaluate the classification system.

1.3 Research methods

This study will use a multi-method approach, structured around the research questions and objectives stated in the previous section. The work is performed in the following phases: literature review, thematic and comparative analysis of existing classification systems, iterative development of a classification system using an action research approach.

Literature review helps to establish the current state of classification system utilization in the construction industry and build a foundation for further steps, while answering the first two research questions and informing research objective two. Upon having an overview of classification system utilization and the attributes related to classification systems, thematic analysis of existing widely used classification systems is performed. In the thematic analysis part of the research each selected system is examined through a prism of several aspects such as foundational logic, scope, depth and interoperability, which helps achieve research objective one. The thematic analysis also informs further investigation into the research question four by overviewing frameworks that align classification with product or material attributes. The information from the analysis is then synthesized into a matrix of attributes of the reviewed systems to be used in the comparative analysis. The synthesis informs research into the research question three and research objective three by taking note of existing practices and known and already addressed demands. The comparative analysis identifies gaps that existing classification systems may have, gathers best practices and builds the foundation for harmonization steps.

The classification system is aimed to be developed from the perspective of designers through the action research approach. This includes identifying the steps necessary for harmonization, starting from creating an initial version with a limited scope. The first version receives feedback, after which it is refined and the scope of the classification system is widened, covering a bigger set of construction products and elements. Input from persons within the industry is gathered to collect opinions on improvements of the developed system and gaps in implementations of existing classifications, together with the previous steps of the research and the development of the classification system research questions five and six are targeted. The output of this phase informs the developments on completing the research objective four. At the end of the project an overview of the entire process is drawn out, completing the research objective five.

2 Literature review

2.1 Classification systems in construction

2.1.1 Overview

As widely acknowledged in the construction field, classification systems play an important role in organizing and structuring the vast amount of information generated throughout construction projects. By providing a consistent framework of categories and indexes or codes, they unify the terminology across various stakeholders, such as architects, engineers, contractors, and facility managers. Thus, each party references the same concepts and assets in the same way. Through this unification, project information can flow across the entire lifecycle of a building, supporting lifecycle estimates and strategic planning. From a practical perspective, classification system adoption benefits cost estimation, scheduling, procurement, risk management, and more efficient resource allocation.

Nowadays, with the rise of international projects, process automation, and the expanding amount of data, a unified BIM language used together with a defined classification system further enables project stakeholders to locate and manage important construction information, proving the increasing importance of classification systems (Pupeikis *et al.* 2022, p. 2).

Furthermore, as noted by Jackson (2020, p. 17), although modern technologies allow for handling large sets of digital data, with potential pattern recognition on a scale beyond reasonable human capabilities, there is still a strong need for a coherent labelling and organizational framework for objects, their properties, and relationships.

2.1.2 Design phase

During the design stage of a construction project, classification systems can serve as a foundation for consistent information management and incremental development of the design. By utilizing standardized categories or codes to building elements, design teams carry the project through various design stages and information captured by the classification index or code is consistent from conceptual design to detailed assemblies with specifications and performance requirements.

According to Ekholm (1996, p. 18), classifications provide utility in shape-based attributes in early design sketches, whereas in later stages there is demand for more refined grouping based on function or assembly

structure, represented by designed elements or work sections. By assigning standardized codes to each design object, designers can incrementally introduce new properties, such as fire resistance or sound insulation, but still keep the clarity on how each component fits into the broader model.

Nowadays classification systems function as systematic means of assigning BIM information to distinct groups. Among the utility that classification systems bring is improved communication and coordination among stakeholders, increased productivity and efficiency of design, interoperability between various software and platforms, reduced errors or omissions (Villagomez & Diamond 2023, p. 8). What is more, as information becomes more easily manageable, better-quality data can go downstream to other stages of construction with cost savings and more sustainable outcomes (Villagomez & Diamond 2023, p. 8). Moreover, the ability to reference external information sources directly from BIM objects adds additional utility (Choy 2015, p. 2). This further reduces the amount of "bloat" in a model and enables access to the right information at the right time. By linking external databases or documentation designers and other stakeholders can quickly access detailed performance specifications or other documents.

2.1.3 Beyond design

Classification utility extends beyond into procurement, construction, commissioning, and facility management. Across these stages, the consistent use of classification systems enables clearer information flow, better coordination, and improved digital workflows. These systems serve not only as shared vocabularies, but also as a framework for the organization of data across tools and disciplines.

During the procurement stage, classification systems play an important role in cost planning, quantity take-off, and tendering. Documents such as Bills of Quantities (BOQ) and pricing schedules can rely on standard classification codes to define project scope. In Finland, for example, the Talo 2000 system is used to generate elemental BOQ and cost plans, ensuring consistent structure from design documentation to contractor pricing (Rakennustieto, p. 2). This can be further used in digital estimating tools, using classifications as anchors for automating cost aggregation.

In the construction phase, classifications also help structure the way tasks, materials, and resources are managed. Contractors frequently adopt the same classification codes from the design and tendering stages to form a Work Breakdown Structure (WBS). This structure supports task planning, scheduling, and cost control on-site. For example, Talo 2000 tables are used in Finland to group work packages and track construction progress

(Rakennustieto, p. 2). Classification systems like MasterFormat also facilitate change order logging by standardizing how work is organized across teams (CSI 2024). Advanced systems like Uniclass 2015 extend classification to temporary works and site resources, offering codes for scaffolding, safety equipment, and construction infrastructure (NBS 2022). These allow digital tools such as field management apps or 4D/5D BIM platforms to assign activities and costs to specific building systems or assets. Afterwards, the information can be transferred into commissioning and handover, and subsequently, into facility management and used in as-built models and digital twins.

2.2 Information flow

Information is an integral part of construction projects, flowing through all of the stages of the asset life cycle. Focusing on construction product and element information, at each phase of a project critical information must be exchanged between various actors, such as architects, engineers, contractors, suppliers, etc., through drawings, specifications, schedules and other. Decisions made on one stage of a project are based on information from previous stages, regulated by the quality of the information flow. Lack of continuous information flow results in fragmentation, which has been acknowledged as a very common issue in the construction industry. One outcome of the fragmentation is manifested in designer decisions getting misinterpreted and procurement orders not matching the designed requirements.

Using classification significantly improves the flow of information between project actors by reducing fragmentation of data. When all stakeholders classify information consistently, it becomes much easier to compile and transfer that information from one phase to the next. For example, during procurement, contractors can take a classified materials list from designers and automatically aggregate items by category for ordering or pricing, rather than manually interpreting each item's description. During construction, progress tracking can be organized by classified work packages or building zones, making it clear which trades or elements the reported information is related to. Modern research by Alhava et al. (2024) shows that by standardizing and harmonizing the data content across design, procurement, delivery and construction phases the disjointed information flow can be overcome. In their proposed framework the metadata definitions are aligned with downstream requirements, enabling automated enrichment and data transfer.

Modern workflows use classification systems at the core of information management for efficient data flow in Common Data Environments (CDE). ISO (2019) defines best practices for information management in BIM projects and specifies classification as a part of requirements for

information organization. It states that digital information container such as models, drawings, or documents should include classification data to enable structured, automated, and interoperable communication between project participants. A pillar of seamless information flow in modern workflows is the use of standard open data schemas like IFC (Industry Foundation Classes). IFC is an information sharing standard developed by buildingSMART (buildingSMART International 2023), it provides a common language for sharing BIM data at the schema level. However, IFC provides only higher level terms that require further extension or inclusion of additional properties (buildingSMART International) that are provided by specific classification systems. Each IFC object, such as IfcWall or IfcDoor can be linked to external classification references, and in practice BIM authors often assign a classification to objects as properties. The integration has significant benefits, therefore efforts have been made to align classification systems like Uni-class with the IFC schema (NBS 2022). In such way classification systems provide the taxonomy for information and IFC provides the structure of information exchange.

While many industry practices limit the structured information flow primarily to early project phases, standards such as COBie (Construction Operations Building Information Exchange) aim to extend this structure into the later stages of a project, particularly at handover. COBie captures essential asset data from BIM in a tabular format and acts as a user-friendly interface for facility managers and owners after the construction concludes (Nisbet). It is typically used to deliver finalized asset information, however, some authors have proposed expanding its use. For example, Hamil (2018) views it as a framework capable of supporting structured information exchange throughout the entire project lifecycle. From this perspective, COBie can function as a persistent classification-linked data format that enhances traceability and continuity across all stakeholders and stages.

Achieving an integrated and seamless information flow throughout the building lifecycle presents significant challenges: handling static versus dynamic information, embedding existing data into new models, and accommodating varied requirements of different developers (Raitviir & Lill 2024, p. 13). Classification systems can mitigate some of these challenges by offering a standardized framework for reincorporating legacy data by marking unused code positions, tracking deprecated labels, and mapping old schemas to new hierarchies, thus preserving backward compatibility and ensuring that no information is left behind as the system evolves.

2.3 Standards and interoperability

Classification systems themselves are anchors of the common ontology of defining how components and processes are categorized, having an impact on how the common understanding between project actors is created (Ekholm 2005, p.1). Standards both create a common basis for classification system creation and provide physical means of data sharing, enabling interoperability.

In this context, ISO 12006-2 is a foundational international standard that defines a framework for classification of construction information and serves as reference for developing compatible classification systems (ISO 2015). ISO 12006-2 does not provide a classification taxonomy itself, but presents principles, common terminology, and a breakdown of structures that can be adopted by national or project -based classification systems. As several major systems have been developed under these guidelines, construction elements like walls classified in one system should in principle be mappable to the equivalent entity in another system. As a bridge between systems the ISO 12006-3 standard provides an extensible taxonomy model, a data dictionary with a framework for standardized object definitions and metadata (ISO 2021).

The ISO 19650 standards address information management in BIM across the project lifecycle. It provides a standardized process for organizing and exchanging information, focusing on use of consistent metadata, naming conventions and data structures (ISO 2018). The concept of a Common Data Environment (CDE) is introduced and requires each information container to have a unique identifier and follow a standard naming schema. An important part of the naming schema is the inclusion of a classification code and a status code indicating the content and approved use of the file (Autodesk). Implementors of ISO 19650 use a national classification or standard to fulfil the requirement. To summarize, ISO 19650 links process standards with classification and data format standards for interoperability. The classification provides the taxonomy for organizing project information and ISO 19650 defines when and how that information is managed and shared.

ISO 81346-1, ISO 81346-2, and ISO 81346-12 define a reference designation system for structuring technical systems across different industries. Part 1 establishes universal principles for breaking down systems into hierarchical subsystems and components, and it introduces aspects such as functional, product, and location (ISO 2018). ISO 81346-2 provides a generic classification of object classes with letter codes based on function for various engineering fields (ISO 2019). ISO 81346-12 adapts the principles to the construction sector and organizes building elements as systems or subsystems in a unified hierarchy (ISO 2022). The standards are used widely in industrial contexts, and see growing implementation in construction, with some

classification systems like CoClass being created based on these standards for more reliable interoperability. However, from the perspective of classifying construction products and elements with the designer utility in mind there are limitations to these standards. Functional roles and stable system identifiers are emphasized, instead of a multifaceted taxonomy. Even with the extension ISO 81346-12, the standards are not typically used in design workflows due to lacking representation of properties, materials or other information required for design or specification. Designers usually rely on ISO 12006-2 based classifications or proprietary systems and use them within schemas like IFC. In practice, an architectural wall or a piece of equipment would get a designation from ISO 81346 to indicate its place in the overall system breakdown, but the designer would still reference another classification scheme for detailed product categorization.

The buildingSMART Data Dictionary (bSDD) is a semantic library based on ISO 12006-3 that provides a unified reference for construction terminology (McPartland). bSDD serves as a centralized hub where different classification standards are stored and interlinked via common definitions (buildingSMART International). This interlinking enables integration and potential translation from one classification to another. The dictionary has been developed as a bridge between different classification standards and data sharing processes. In practice, however, bSDD's implementation has been uneven: its interface and search capabilities are not optimized for designers seeking to discover or compare similar products, making it seldom used as a day-to-day tool. Nevertheless, the underlying mappings and cross-reference tables it maintains are a good the interoperability foundation.

As mentioned earlier, information management standards like IFC and COBie enable data exchange across software platforms and project phases, but they do not themselves define how building elements or products are categorized. IFC provides a neutral schema for sharing geometry and semantic properties, and COBie prescribes a tabular deliverable for asset hand-over or other use cases.

2.4 Harmonization and designer benefits

In the context of construction information management, harmonization refers to standardizing data structures for all participants to interpret and use information consistently. Classification systems can serve as one of the initial tools for creating a common organization for construction product and element data. The consequent harmonization results in more efficient collaborative work with coherence and compatibility of data (Lima et al. 2021, p.2). When a classification system is harmonized, its codes, hierarchies, attributes and definitions are consistently applied and it becomes inherently

interoperable. However, within a single classification system harmonization can be achieved by following uniform logic, delimiting conventions, clear definition of hierarchies and their levels, standard metadata fields, well selected terminology and relationships between interrelated elements.

From the designer's perspective, apart from information flow continuity, harmonized information brings definite advantages, as when construction product and element data is classified uniformly, designers can discover and compare relevant items. For the purpose of this study, "product discovery" is defined as the process by which a designer searches, identifies, retrieves, and evaluates construction products and elements in a systematic way. Product discovery relies on a harmonized taxonomy, enabling filtering catalogues by function, material, performance, etc. The harmonized information subsequently allows for information comparison and object insertion and enrichment in design processes. For example, digital catalogues and BIM libraries that are tagged using a classification schema shall allow quick searching by function or attribute. In building models this results in every object having the same set of data fields, as noted by Watson D. (2018) using a common classification "will result in BIM objects and specifications having a consistent structure". However, expanding this idea further to product discovery would mean that queries could be made for finding equivalent or similar products or elements.

2.5 Review of existing classification systems

Before establishing a new classification framework, it is essential to understand how existing systems are structured, applied, and compared in both research and practice. This context offers a foundation for identifying limitations and gaps that a new solution might address. In the thesis by Knopp-Trendafilova (2010), a comparison between classification systems is carried out in an attempt to find a link between the systems and structural models, where a basic overview is given on most widely used classification systems in Finland and internationally. It is emphasized that there is a strong need for a common ground between classification systems for interoperability and integration into the BIM workflows. Other comparisons have been carried out various authors, including Pupekis et al. (2022), where CCI is compared against Uniclass 2015 in a generalistic SWOT analysis. Another comparative study by Afsari et al. (2016) closer relates to the questions raised in this paper with a more descriptive review of classification systems. Nevertheless, the comparisons have been made from different perspectives, not aligning with the goals of this study, therefore a closer look is required to answer the questions of this research.

The classification systems in construction have evolved to meet various functional, organisational and regional needs. They serve to organize information about building components, products and processes or activities to help create a more seamless information flow throughout the construction process and improve communication between the parties involved in a given project. In choosing which systems to review, priority was given to those most widely used in major markets and recognised by practitioners: Talo 2000 and LVI Tuoteosa in Finland; Uniclass in the United Kingdom; ETIM across Europe; UniFormat, MasterFormat and OmniClass across the United States and Canada; DIN 276 in Germany; CoClass in Sweden; and leading lifecycle or performance-based tools (LCA-based databases, SKOL ry guidelines) in the EU.

Most of the classification systems either are based on or have connections to the ISO 12006-2 framework, that provides a structured way of organizing construction information. Despite this, each classification system tends to differ in accordance with specific challenges that it is designed to solve and with the foundational idea behind each.

A review of several commonly used national and international classification systems around the world has been undertaken in the following subsections. The focus of the review is to gain a deeper understanding of the foundational ideas behind each system and what problems they aim to solve. There is a look into the organizational structure of each system and into how they facilitate information transfer across various stages of construction. Afterwards, a comparison is given with emphasis on the strengths and weaknesses of each system from the perspective of a designer. Each classification system in this section is examined to understand the rationale behind it and its scope. In general, they all serve to contribute to the broader goal of improving the flow of information through a project's lifecycle. This section will highlight the following aspects of each classification system:

- The problem they solve;
- The depth and organizational structure;
- The foundational idea;
- How they facilitate information transfer.

As many of the systems are based on or linked to ISO 12006-2, clarifying the role and scope of this reference framework can be beneficial. According to the International Organization for Standardization (ISO) official website, ISO 12006-2:2015 is providing a number of recommendations for organizing construction information, guidelines for classification table titles, standard definitions and recommended structuring principles. According to ISO, if separate classification systems are created in accordance with

ISO 12006-2:2015, then they will be harmonized and can be linked together. It is also important to acknowledge that the framework is designed to serve the full lifecycle of construction projects, from briefing to demolition (ISO 2015).

2.5.1 Talo 2000

The Finnish Talo 2000 project classification, also referred to as Building 2000 or Construction 2000 “covers building elements, building services elements, structural parts of the former two, as well as project, property management and user tasks” (Construction 2000 Classification Committee; Haahtela-kehitys Oy; Rakennustietosäätiö RTS 2010). The classification of “building elements” and “services elements” is given more particular focus in the project classification.

Talo 2000 serves designers by providing a clear and organized table of element groups, treating the integrated approach of viewing design, costs and construction planning. The system takes into account to also organize quantity measurement rules for each building element, empowering information flow. According to the paper The Finnish Construction 2000 Classification System (Rakennustieto n.d.), the classification tables are applied at various stages of construction projects. They are used for design briefs, cost planning, and contract details; for BIM integration and specifications; for cost estimates, budgeting, and on-site planning; and for procurement and production oversight.

This classification system solves issues of a lack of a common language between different stakeholders, inconsistencies in measurements and specification methods, fragmentation in data flow between design and cost management, difficulties of integration into BIM and specifications, and it improves guidance for decision making in the design phase.

In section 1 Building elements Talo 2000 takes a perspective on a building as a set of coherent assemblies, each of which has its own functional identity. The classification is organized in element groups with subcategories. For example, a roof is considered to be its own self-contained system including not only the structural elements but also accessories and components that are relevant to the roof functionality and maintenance. Thus, roof safety products are under the same category together with skylights and roof substructures. Each category has its own code and definition to help with information flow from early phases of design to procurement and maintenance.

In addition, Talo 2000 includes a product nomenclature system for classifying products that are permanently installed in a building or used

during construction (Rakennustietosäätiö RTS sr 2000). The product nomenclature is used for identifying products in public directories, procurement and documentation. This system largely follows the same principle of functional approach.

2.5.2 Uniclass

The British Uniclass classification system provides a comprehensive categorization of every aspect of the built environment (NBS 2022) and is mentioned in ISO 19650-2 National Annex as a tool for classifying information containers and their contents. It covers everything from large complexes to individual products, and has a total of 14 classification tables. The tables are representing assets like elements/functions (EF), systems (Ss), products (Pr), Materials (Ma) and others.

By grouping classifications across design, specification and management, Uniclass provides a standardized way to label and manage various components of a project. It solves many problems and needs in regards to information flow through every phase of a project with a unique way of labelling each asset type. From the point of view of designers, Uniclass offers a logical code system that can be integrated directly into BIM and other tools. This provides a unified way of annotating project documentation.

Uniclass is also regularly maintained and updated for the purpose of keeping the system flexible and able to include newer developments like new products, tasks or other.

Each table of this system has several levels of hierarchy, providing a very deep and detailed categorization. This sophisticated organization is also achieved through a vast number of tables, resulting from the task of covering every asset in a construction project. As an example, a complex item as a roof is represented in the EF table as a functional entity like “Flat roofs”, in the Ss table a roof structure type can be found, and in the Pr table are located individual roof products.

The Uniclass tables are integrated with BIM tools and a code is applied to specific model elements, with the relationship between Uniclass and IFC being emphasized (NBS 2022). Thus, the element can be tracked in stages of design, on specification platforms and even in facility management platforms.

It is important to note that for the need of finding products or BIM objects on the NBS Source website however, a different categorization of items is used, that is based on the Uniclass. This might be done due to the

fact that the Uniclass is a massive system which is not primarily designed for the needs of the designers.

2.5.3 UniFormat

UniFormat, developed by Construction Specification Institute (CSI) standardized classification system is used throughout the construction industry of the United States of America and Canada. It organizes information about building elements and systems based on functional approach. The About UniFormat page on CSI's official website (CSI) places additional emphasis on early design stages and cost estimation as the primary application of the system.

The system improves communication between design teams and cost estimators by focusing on grouping assets by function. With such approach it simplifies comparison of design options and conduction of early budget estimates.

The table includes three levels: Major Group Elements, Group Elements, and Individual Elements. The depth of the system makes it straightforward to manage, however some detail is sacrificed. Nevertheless, the system fully serves its purpose in the initial phases of a project, where details can often be vague. Furthermore, as the project is proceeding, the classification can be mapped to more detailed systems like MasterFormat for purposes of specifications and other down the lifecycle.

The foundational approach of grouping items is in each respective function, such as "Substructure", "Shell", "Interior", etc. These are considered Major Group Elements. Therefore, as an example, roof related items would be found under separate groups of level 3 like "Roof Construction" and "Roof Openings", tracing paths back to their respective level 2 groups and back to level 1 group "Shell".

2.5.4 MasterFormat

MasterFormat is developed by the same organization as UniFormat, CSI. It is widely used across North America. According to CSI's official website (CSI), MasterFormat tables are designed to be used through all of the stages of a construction process, with emphasis on specifications and organization of construction documents.

MasterFormat is a more detailed system than UniFormat and it is developed by organizing construction assets by trades or work results. As such, the system makes clear scopes of activities and items that reduces

inconsistencies and miscommunication, with each category containing elements grouped by the required works to be completed. In such way, the framework is aligned with procurement, contracts and the process of construction itself, while being less relevant for the requirements of BIM workflows. Nevertheless, MasterFormat is a powerful tool for information transfer, as its codes can be linked to BIM objects and cost databases.

The structure of the system is quite detailed, containing 50 Divisions that have sublevel specifications of items. As an indicator of how the system is configured, a similar example of how roof related elements or items are classified can be considered. Each particular part of a roof structure can be found under a general group that is defined by the nature of work required and a general source material group.

2.5.5 OmniClass

The third system developed by CSI and used across North America, OmniClass is organizing the built environment from several perspectives like products, spaces, phases, etc. OmniClass has 15 interrelated tables that incorporate MasterFormat and UniFormat components. It is important to note that Table 21 “Elements” is based on ISO 12006-2 reference framework.

The classification system addresses the need to provide value to all stakeholders and project stages with its several tables. Therefore, it has been developed additionally to support designers and BIM workflows. As per the official website (CSI), OmniClass is intrinsically designed for organization of BIM objects and Revit families in mind, with information flow being supported further into facility management and building lifecycle data.

With the multifaceted nature, the classification system uses functional approach of defining element groups in Table 21 “Elements” and product based approach in Table 23 “Products”. Tables have significant level of depth, making them suitable for interoperability.

Thus, in Table 21 roof related elements would be found in functional groups under level 1 “Shell” with a reference code in Table 22. In Table 23 they are grouped by the product type, that can be a single element or an assembly, based on manufacturing and distribution. Table 22 on the other hand may represent the same elements through the lens of tasks performed on site. The relationship between each code or item from each of these tables then can be established by users.

2.5.6 DIN 276

DIN 276 is a German standard that has been developed for organizing assets under construction cost groups. The standard provides a set of classifications with codes to track costs from budgeting to final accounting. It has been designed to increase transparency and enable straightforward separation of cost types into categories (DATAflor).

The standard's degree of sophistication is represented in 3 levels. These levels are rather wide cost groups that include general parts of buildings or other items like terrain works or plants. DIN 276 primarily achieves consistency in terminology and cost grouping, however it can be linked to other specifications.

2.5.7 CoClass

CoClass is a modern Swedish classification system that was designed to replace older national standards such as BSAB 96. BSAB 96 was intended for use in only from design to production phases and was not sufficient for the entire built environment. CoClass has been developed with specific goals in mind to allow information flow and management through the entire lifecycle, to be accessible with modern workflows and to have scope and depth suitable for all assets of built environment (Svensk Byggtjänst). It is offering consistent terminology across various phases like design, construction process and facility management.

A unique aspect of CoClass is its maintenance and keeping up to date, as it is an online system of tables.

The system provides a flexible multifaceted approach to grouping items and creating unique but standardized classifications, virtually, for each project. CoClass provides a framework of classifications with relatively open ended final levels, left to be linked by users. With the use of API, however, users can create workflows suitable for them and still have the benefits of the information flow being consistent. The codes offered by the system are inserted into BIM models and with the help of auxiliary tools information about materials, structural systems, tasks and other is tracked down the lifecycle.

As an example, roof elements may be grouped under different categories of level 1, which complement each other for the respective fields. This flexibility allows for a significantly higher level of usability for all involved specialists, however comes at a cost in requiring additional work to make use of the functionality.

2.5.8 LCA – lifecycle-based classification

Life Cycle Assessment (LCA) tools often organize product data under categories based on main material origin or product type, where each entry has assigned environmental impact indicators such as global warming potential, energy use, and water consumption. Systems such as ÖKOBAUDAT, CO2data, One Click LCA, and others compile databases in such way, resulting in the ability to differentiate materials and products on the basis of quantifiable lifecycle metrics.

From the perspective of designers, this approach brings a way to compare products based on environmental impact and track information up until the demolition of a building or a structure. Additional utility is provided by Environmental Product Declarations (EPD) and European standards like EN 15804, which can result in a feed of information into the databases, linking categories with relevant data.

2.5.9 SKOL – Property-based product specification guideline

In accordance with the European CE marking, manufacturers issue a Declaration of Performance (DoP) that is listing the key or characteristic properties of a construction product. Designers can set performance thresholds such as, for example, compressive strength, fire rating or thermal conductivity for a project and find a suitable product accordingly.

SKOL ry (Suunnittelu- ja konsulttitoimistojen liitto) has published Rakennetyyppien Yleiset tuotemäärittelyohjeet (General product specification guidelines for structural types) in Finnish that organizes construction products according to characteristic properties grounded in European CE marking requirements. A table drawn from SKOL's paper (Appendix 4) illustrates how these categories are structured in a hierarchy and includes English translations.

According to Rakennustieto, the work has been carried out to create a way of product specification based on properties for an improved ability to find equivalent substitute products or solutions. From this approach designers benefit from a transparent basis for compliance with regulatory requirements. By specifying items by their technical performance, the benefit of flexibility and cost-efficiency is achieved also in procurement, as equivalent or superior products can be substituted without downgrading the quality of the project. Overall, property or performance-based classification approach can lead to a more unified language for communication between designers, contractors and suppliers.

2.5.10 LVI Tuoteosa (Version 1.0)

The LVI Tuoteosa classification system has been created specifically for classifying heating, water and ventilation components in Finland and is provided within the Reference Data tool on the Suomi.fi website (Koodistot). The system has been introduced to standardize naming and identification of HVAC and plumbing components in BIM.

The system has 2-3 levels of depth in its hierarchies, however it is presented as a list of component types with codes and names. It has been developed with machine readability in mind. The coverage of LVI Tuoteosa is aimed to be extensive. Items are grouped by functional elements or assemblies.

Designers embed LVI Tuoteosa codes as user-defined attributes in their BIM models and on symbolic drawings. During design coordination or automated compliance checks, the presence and correct classification of each component can be verified by rule-based software. The industrialized nature of HVAC and plumbing products differs LVI Tuoteosa's highly defined approach from classification systems designed for other disciplines. Parts are already standardized by manufacturers, therefore designers benefit from a classification that mirrors the product catalogues they use.

2.5.11 ETIM – Attribute-based classification

ETIM has been developed as an internationally viable classification system (ETIM International), starting from the electrical sector and expanding the coverage to other construction products. The system currently covers HVAC, plumbing, building materials, tools and other. The aim is to solve inconsistent product data, different terminologies, inconsistent units and product descriptions.

ETIM has 2 levels of hierarchy with product "Groups" and "Classes" (ETIM International), where the tables are otherwise extensively covering products. Every product class carries a unique ETIM code and a prescribed set of attributes, which may be numeric or textual. Each attribute is bound to a specific unit of measurement or list of permitted values for preserving the data harmonized.

The foundational concept behind ETIM is feature-based standardization. By defining which attributes are relevant for each product class, the system allows engineers and architects to filter and compare products on the criteria they are interested in. Industry consensus is maintained through ETIM International, a non-profit body that compiles input from national

chapters and periodically updates the classification to accommodate new technologies.

ETIM facilitates information transfer in multiple ways. Manufacturers publish their catalogs in ETIM-compliant formats, which can be imported directly into BIM systems. For BIM workflows, the newer ETIM Modelling Classes link each product class to IFC object types and property sets, enriching 3D models with standardized product data that downstream tools for quantity take-off, procurement modules and facility management can receive and process automatically.

Because HVAC and other MEP products are typically industrially manufactured to tight specifications, ETIM's attribute-driven approach fits well. Designers know that each feature corresponds to a real, measurable property in the factory-made item.

2.6 Comparison of classification systems

After reviewing the various classification systems that have been developed in different regions and for different tasks and scopes, it has become clear that each uses a different foundational logic. In the discussed classifications, distinct groups of foundational ideas or approaches have been observed:

1. Functional assemblies;
2. Trade-based, Cost-based;
3. Based on material or environmental impact;
4. Multifaceted approach;
5. Performance attribute-based.

The differences between each approach reflect the context in which the systems were created, often local or specific to the task. Each ideology comes with a set of advantages, however brings forward the challenge of scope and depth. A more inclusive system has a risk of creating too much complexity, while a narrower and less cumbersome system might miss important groups of assets or perspectives needed for effective data flow. Both the context and the foundational idea result in differences in other aspects of a system. Table 1 below compares the classification systems according to their foundational idea, scope and depth, general utility, and utility from the perspective of a designer. Scope indicates how broad is the coverage of a system, and depth relates to the number of hierarchical levels or tables. General utility captures the system's principal applications such as cost estimating, specification writing, or sustainability assessment, while designer perspective evaluates how readily the system supports everyday design tasks like product discovery, documentation, and decision making.

Table 1. Comparison of reviewed classification systems.

Classification system	Foundational idea	Scope	Depth	Utility	Designer Perspective	Region / Country
Talo 2000	Functional elements or assemblies	Moderate scope with element groups	Moderate depth, up to 4 levels	Integrates design, cost, measurement rules	Functional and holistic view of building parts	Finland
Uniclass	Multi-table	Comprehensive scope, 14 tables	High depth up to 5 levels or more	Data flow from design to facility management, complex asset coverage	Detailed but can be difficult to navigate	United Kingdom
UniFormat	Function-based	Moderate scope	Moderate depth, up to 3 levels	Early design, cost estimating	Useful for conceptual design but not detailed for further use	USA, Canada
MasterFormat	Work results/trade-based	Wide scope	50 divisions with sublevels	Specifications, procurement, contracts	Not focused for designers	USA, Canada
OmniClass	Multifaceted	Comprehensive scope with 15 tables of interperspective grouping	Varying	BIM integration, multifaceted grouping to serve virtually all stakeholders	Flexible and comprehensive, however requiring cross-referencing	North America
DIN 276	Cost groups	Moderate scope	Moderate depth with up to 3 levels	Budgeting	Not focused for designers	Germany
CoClass	Flexible, object-oriented	Flexible	Flexible, multilevel and multitable	Configurable and versatile for data flow but requires effort to implement.	Flexible and configurable through API, high potential utility	Sweden
LCA-based (e.g., co2data)	Based on material, environmental impact	In general scope is focused on material type	Extensive depth in some cases.	LCA, data flow	Optimal for LCA, limited scope	International (tool specific)
SKOL ry - General product specification guidelines for structural types	CE characteristic properties	Focused scope	Moderate depth with up to 4 levels	Performance based grouping	Easier product and substitute search by performance thresholds	Finland
LVI Tuoteosa (v1.0)	Product/component-based	Around 500 HVAC & plumbing component types	Shallow hierarchy, flat list	Machine-readable BIM tagging, automated compliance checks	Mirrors manufacturer catalogs; precision for MEP design	Finland
ETIM	Feature-based product standardization	Mostly MEP items and attributes	Two-level hierarchy (Product Groups and Product Classes) + attributes	Unified product data exchange, BIM integration	Highly precise attribute filtering; learning curve for novices	International (EU)

As mentioned above, each system is created based on a specific logic or idea, whether functional, trade-based or otherwise, leading to differences in other aspects. Broadly speaking, these logics fall into five categories: functional assemblies (e.g. Talo 2000, UniFormat), which group items by their role in the building; trade- or work-result orientations (MasterFormat, DIN 276), which mirror contracting and cost structures; material or environmental impact perspectives (LCA-based tools), which organize data according to sustainability metrics; multifaceted schemas (Uniclass, OmniClass, CoClass) that layer element, product, space and activity tables; and product-centric, attribute-driven approaches (ETIM, LVI Tuoteosa), which present a flat list of classes enriched with standardized technical parameters.

Comparing scope and depth, systems offer varying coverage widths, degrees of sophistication and levels of hierarchical structuring. For example, Talo 2000 and UniFormat both offer moderate depth that is adequate for early design and basic cost estimation, whereas Uniclass and OmniClass reach a significant depth in addition to wide table coverage to classify every asset type from initial design to facility management. MasterFormat has a wide scope covering work result groups in its 50 divisions, thus making it suitable for procurement and specifications, but less so for design workflows. On the other hand, a classification system such as CoClass is distinctly different from others by providing a flexible multilevel table organization, virtually fitting any stakeholder's need, with a reliance on online services and API integration, however this utility comes at the cost of expending extra resources on adopting it and more complex cross-system mapping. At the opposite end of the hierarchy spectrum, ETIM and LVI Tuoteosa offer very shallow structures, ETIM with just two levels plus rich feature attributes, and LVI Tuoteosa as a flat list of MEP component types prioritizing product precision.

From a lifecycle perspective, some systems clearly align with particular project stages: UniFormat and Talo 2000 excel in schematic design and early cost planning, while Uniclass, OmniClass and CoClass are better suited to later stages, including facility management and operations. ETIM and LVI Tuoteosa, in contrast, are optimized for the detailed selection and specification of manufactured components in MEP design and procurement.

The systems also differ in their utility and intended stage of application. MasterFormat, for example, is tailored to improve specification creation, procurement, and contracts, while on the other hand UniFormat is often used for early design and early cost estimation. Talo 2000 provides good coordination between design and quantity take-off with the help of its link of building elements and measurement rules. DIN 276 is very valuable for budgeting with its focus on cost grouping, and LCA systems focus on

prioritizing environmental impact metrics and realize efficient information transfer for sustainability assessment and tracking. ETIM and LVI Tuoteosa provide value for product catalogs, automated BOM extraction, and compliance checking at later stages.

Interoperability and the ease of mapping between systems is an important practical consideration. Flat, code-driven schemas such as LVI Tuoteosa can be mapped to elemental systems like Talo 2000 with relatively simple look-up tables, whereas mapping multi-table systems like from Uniclass to MasterFormat require more extensive cross-reference tables and have a risk of semantic drifting in the mapping.

Viewing the systems from a perspective of the needs of a designer, each system offers varying levels of utility and usability. Talo 2000 has a straightforward division by functional assemblies that fits well with how a designer would view the project. Uniclass and OmniClass are comprehensive and provide good structuring for clear information flow, however they might be difficult to navigate with their high degree of sophistication and require cross-referencing. MasterFormat and DIN 276 have great value for other specialists, but are not focused on providing utility for designers. LCA systems have a niche and are valuable in design, however, are a separate workflow from usual structural, architectural or other design. Performance-based approach developed by SKOL ry has potential in being an organizational structure that has great value for designers, as by grouping products based on CE marked characteristic performance it enables easy product search and substitution with regulatory compliance, fitting needs of structural designers or other professionals dealing with performance thresholds. Finally, ETIM and LVI Tuoteosa are especially useful to MEP engineers for their precision and direct BIM integration, helping with the demand for attribute connection and machine-readability.

In summary, the reviewed classification systems form a spectrum defined by five core logical approaches: functional assemblies, trade- or cost-based grouping, environmental impact perspective, multifaceted schemas, and product centric attribute approach. At one end of the spectrum elemental and cost-oriented systems such as Talo 2000 and UniFormat deliver moderate hierarchical depth tailored for early design and basic budgeting, while at the other, ETIM and LVI Tuoteosa present very shallow, flat structures optimized for precise MEP product specification and direct BIM integration. Between these poles lie multifaceted frameworks (Uniclass, OmniClass, CoClass) that span multiple tables and support the full project lifecycle design through facility management, but carry extra complexity and mapping overhead. Trade-driven systems like MasterFormat and DIN 276 excel at

specification writing and cost accounting, while LCA-based tools specialize in sustainability metrics.

Building on the tabular comparison, four comparative dimensions were created to reflect on how each system is positioned in the overall context: 1) Depth – hierarchical granularity; 2) Scope – breadth of coverage; 3) Utility for designers – support for product discovery, documentation, decision making; 4) Interoperability – ease of mapping, data exchange and native integration with BIM workflows. Depth and scope are a common reflection of structuring challenges in classification frameworks (see ISO 12006-2). Utility for designers is based on the design process benefits discussed previously. Lastly, interoperability reflects the emphasis on consistent information organization and exchange in project delivery processes (see ISO 19650) and published mapping practices between classification and data schemas (e.g. Uniclass to IFC guidance). These dimensions are plotted in Figure 1 as a radar chart.

For each classification system ranking, depth and scope were read from official tables and documentation and ranked between the selected classification systems. Utility for designers was judged against evidence for support for product discovery, documentation structure, and decision making. Interoperability was assessed based on the presence of published mappings, APIs and references to cross-tool usage.

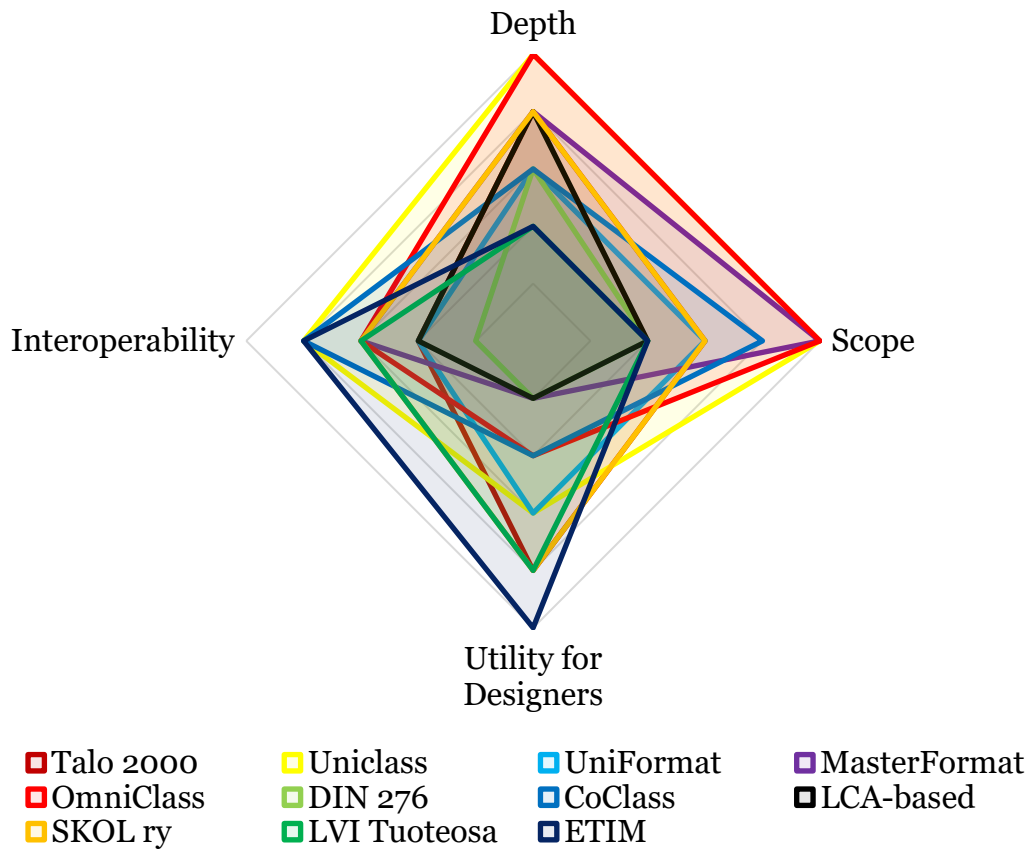


Figure 1. Radar chart comparing reviewed classification systems.

Figure 1 allows a concise visual representation of how each classification system balances the four dimensions, making the gaps more apparent. Most of the reviewed systems fall short in providing utility for designers in product discovery and decision making along with interoperability, while systems that do, are limited in scope and depth and cover items from the more industrialized areas of construction products. The graphical representation shows that no existing system adequately combines all the criteria, proving the need for a harmonized approach to bridge the divides and offer granularity to differentiate products with a user-friendly structure, have broad coverage, and good data exchange capabilities. Element- and cost-based schemes like Talo 2000 and UniFormat simplify early-stage planning but lack the product-level detail required for precise selection. Trade-oriented formats such as MasterFormat excel at procurement but are too granular and construction-focused for conceptual design. Multifaceted frameworks (Uniclass, OmniClass, CoClass) cover the entire project lifecycle but can overwhelm the user with deep hierarchies and impose heavy mapping tasks. Environmental and performance-based tools target sustainability or regulatory compliance but remain a tool for a separate specific purpose and workflows. Product-centric standards (ETIM, LVI Tuoteosa) offer the

precision and machine-readability demanded by MEP engineers, however they are narrowly scoped only to manufactured components and do not integrate seamlessly with functional or trade classifications. Together these gaps make it difficult for designers to quickly find and compare products in the context of their broader assemblies and workflows, while have the information flowing through the phases of a construction project. The gaps suggest a designer-centric system that balances functional grouping, ease of navigation, and standardized attributes for product differentiation, while remaining interoperable with existing cost, trade, and lifecycle schemas.

3 Action research

3.1 Rationale for action research

The method adopted for the development of the classification system is Action Research (AR), with iterative cycles of development, feedback and refinement.

According to Azhar (2007, p. 14) AR can be beneficial in both practical problem solving and in improving competencies and knowledge of researchers and practitioners. The author also states that the method has special utility in research where several disciplines are involved, and investigations of technological changes can be analysed in organizational context. Action research can also be viewed as a process of investigating one's own research in order to verify if it is aligned with the objectives. McNiff & Whitehead (2022, p. 15) note that action research is another name for "researching your own learning" and characterize it as a practical way of viewing and reflecting on own practice. Connaughton & Weller (2013, p. 2) highlight that Kurt Lewin, often credited as a pioneer in AR, described it as a cyclical method of organizational change in which each phase involves planning, taking action, and examining the outcomes of that action. The same authors (2013, p. 3) outline several key features of AR that distinguish it from other methodologies, among which is "a focus on how learning and change processes become self-generating and self-maintaining". Furthermore, Koshy *et al.* (2010, p. 4) underscores the personal and professional development resulting from action research.

The AR approach is well suited for the development of a classification system, as it includes working with various disciplines and allows for continuous refinement and scope adjustment. As the final product of this thesis is aimed to be used by designers, it is of significant value to have the designers requirements represented through feedback.

3.2 Action research model

The stages of AR are not rigidly sequential and may evolve with new acquired insights, this is brought up by Koshy *et al.* (2010, p. 5) in reference to Kemmis's and McTaggart's definition of action research as participatory research involving a spiral of self-reflective cycles. Kemmis and McTaggart (2005, p. 7) present AR as a series of planning, acting and observation, reflection on results, and incorporation of the reflections into the subsequent planning (see Figure 2).

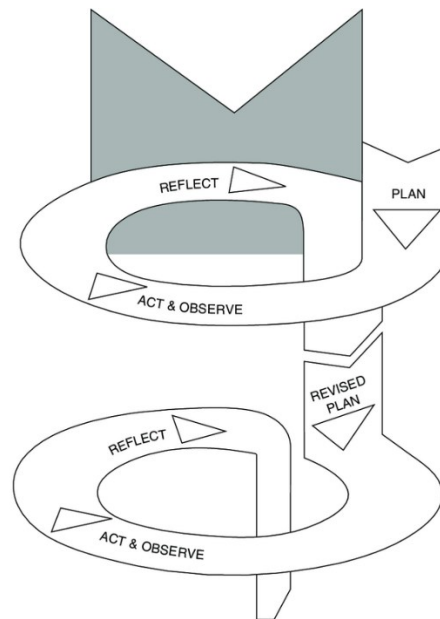


Figure 2. “The Action Research Spiral” (Kemmis & McTaggart)

Figure 2 illustrates the Action Research Spiral, showing how each cycle builds on the previous one. Instead of returning to the same starting point, the spiral continues forward, signifying continuous progress even though the routine work may be still the same. This model of AR is suitable for the current work due to the flexibility it provides in developing and refining a classification system targeted towards designers.

Steps of the first AR cycle (plan – act and observe – reflect):

1. Plan: Synthesize literature and comparative review;
2. Act: Prototype concept creation and explanation, prototype schema and component overview;
3. Observe: Interviews and evaluation of the first version of the prototype;
4. Reflect: Reflect on performed steps, challenges and outcomes from the first iteration and potential for further actions.

Subsequent cycles follow a similar structure, with minor adjustments as the development progressed.

3.3 Data collection and analysis

As the classification system was developed and built up iteratively, this study collected practitioner input in three iterative rounds: semi-structured interviews in Cycle 1, an e-mail survey in Cycle 2, and a final review in Cycle 3. The purpose was to evaluate the developed prototype versions, understand usability issues, and translate feedback into steps necessary for changes.

Nine practitioners selected for interviews and further feedback represented a variety of professional backgrounds, including architecture, structural design, MEP design, BIM coordination. Professional experience was ranging from five to fifteen years. Across the board, respondents reported engagement in multiple project phases, from early concept design to detailed design, and, in many cases, construction or commissioning.

In Cycle 1 nine semi-structured interviews were conducted with practitioners from March to April 2025, some were performed in a face-to-face format, while others via Microsoft Teams online meetings. Interviews lasted from thirty minutes to an hour, and the interviewees' answers were written down as notes. The interviews opened by establishing each participant's background, project involvement and information sharing preferences. Due to not every interviewee agreeing to include their name in the report, all the information gathered from the interviews in this study is anonymous. The interview protocol and the full interviewer notes are listed in Appendix 1.

Cycle 2 and 3 included e-mail-based surveys sent to the same practitioners. The e-mail exchanges were informal by design, including annotated classification prototype versions and a set of prompts. Results arrived as free-form e-mails. Key points were extracted into notes in a similar format as the interview notes to maintain a consistent collection of information. The e-mail prompts and notes are provided in Appendix 2 and Appendix 3.

All interview and e-mail materials were reviewed with systematic marking of recurring issues and suggestions. Similar points were grouped together into themes. Each theme was taken as a concrete change step for the relevant prototype version.

4 Iterative prototyping and evaluation

4.1 Cycle 1

4.1.1 Synthesis and plan

The comparison of existing classification systems highlighted two main issues, namely, the trade-off between depth and usability, and the data exchange between separate schemas. Systems such as Uniclass and OmniClass have exhaustive hierarchies, but can be difficult to navigate and a product-centric system like ETIM has great precision, linked product attributes and has great interoperability, however has a limited scope in building elements and products. Other systems rely on different foundational logics that can be difficult to map and provide good interoperability. From this synthesis emerged the insight that no single foundational approach to creating a classification system can satisfy both the broad scope, granularity for ease of use and precision at the component level. Even so, existing classification systems have wide use, there is a substantial number of systems in use throughout the world, and each has proven its utility. Building on this understanding, the focus shifts from just inventing yet another taxonomy to creating a tool that acts as both a classification system and an interoperability hub. By creating a schema that has links to other systems and enhancing it with consistent attribute data, the tool will be able to make use of the strengths of existing systems while filling the observed gaps.

Rather than attempting to replace existing systems, which already serve their use cases and are widely used, the goal is to develop a complementary classification tool tailored to designers' needs. This tool will emphasize granularity for detailed product identification and integration of product attributes. Recognizing the strengths and established practices within current systems, the approach will focus on enhancing interoperability, enabling users to use existing classification frameworks by connecting them. What is more, the harmonized framework must present a coherent, logical structure and avoid ambiguous or overlapping categories, so that designers can intuitively understand relationships without confusion.

Cycle 1 includes the creation of the prototype concept, outlining the schema and the components of the system, gathering insight and prototype evaluation from practitioners in the form of interviews, and reflection.

4.1.2 Prototype concept

Based on these findings, the first version of the classification system is created not as a static taxonomy, but as a tool with flexible foundational logic with interoperability as one of the embedded characteristics. It combines multiple parallel hierarchies of tags: spatial position, material composition, functional role. While no existing classification system combines these perspectives into a single structure, they are well established in literature and are consistent with structuring principles in ISO 81346-12. ISO 81346-12 distinguishes between functional, location, and product structures, where the former is represented by the material hierarchy in the prototype by focusing on what an object is physically. Each hierarchy is independent, however all of them converge through a tagging mechanism for more flexible searching. This approach is aimed to allow designers to navigate by their preferred perspective.

Each tag has a compact machine readable code that also serves as a human readable label. Combinations of tags are linked to equivalent nodes in Talo 2000 and Uniclass as a key step. This functionality can also allow the system to serve as a translation hub to enable data synchronization between BIM tools, cost databases, etc.

4.1.3 Schema and components

The prototype of the system was represented as several parallel tree structures for the ease of visual understanding. The coverage of currently implemented nodes is not comprehensive and they serve as an example. The spatial hierarchy captures an item's position in a building as viewed from the position in the building's structure, and other general item groups. For both levels of this hierarchy the naming was selected in accordance with Uniclass naming conventions, as it is the largest classification system in the English language utilized in Europe. Level one covers major building zones or systems, e.g. roofs, walls and barriers, structural elements, substructure, lighting and electricity, ventilation, air conditioning and space heating. The spatial hierarchy is represented in Figure 3.

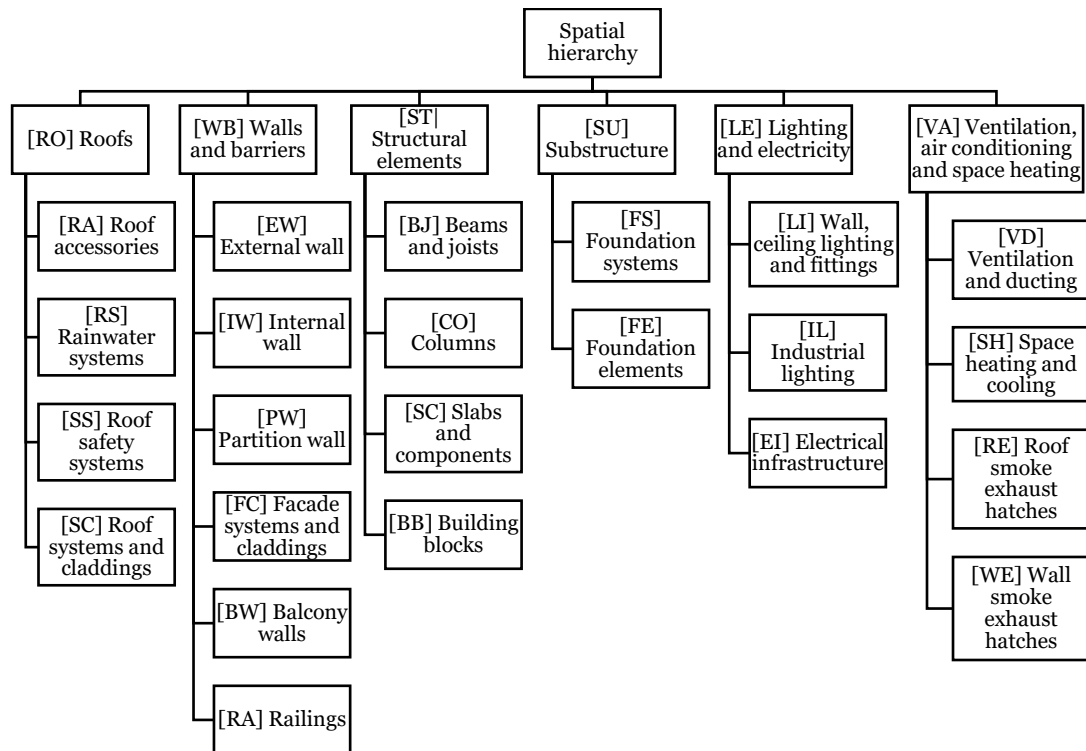


Figure 3. Spatial hierarchy of tags.

The functional hierarchy (Figure 4) continues to draw on the terminology found within Uniclass and includes a singular level with nodes such as waterproofing, sealing, insulation thermal, insulation sound, boards, paint, coatings, and plaster. It is, however, grouping items by their function, thus both the spatial and the functional hierarchy can serve as specifying elements for one another.

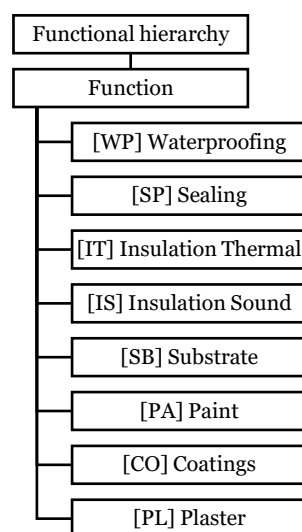


Figure 4. Functional hierarchy of tags.

For the material tags a 3-level hierarchy of tags was created based on the German ÖKOBAUDAT (ÖKOBAUDAT) system (see Figure 5), which is a comprehensive database that classifies building materials and other construction related items based on LCA data. However, the prototype version presented here is not comprehensive, as it includes only a limited selection of categories and does not yet cover the full range of material types in the source system. The material hierarchy is intended to categorize an item by its characteristic material composition.

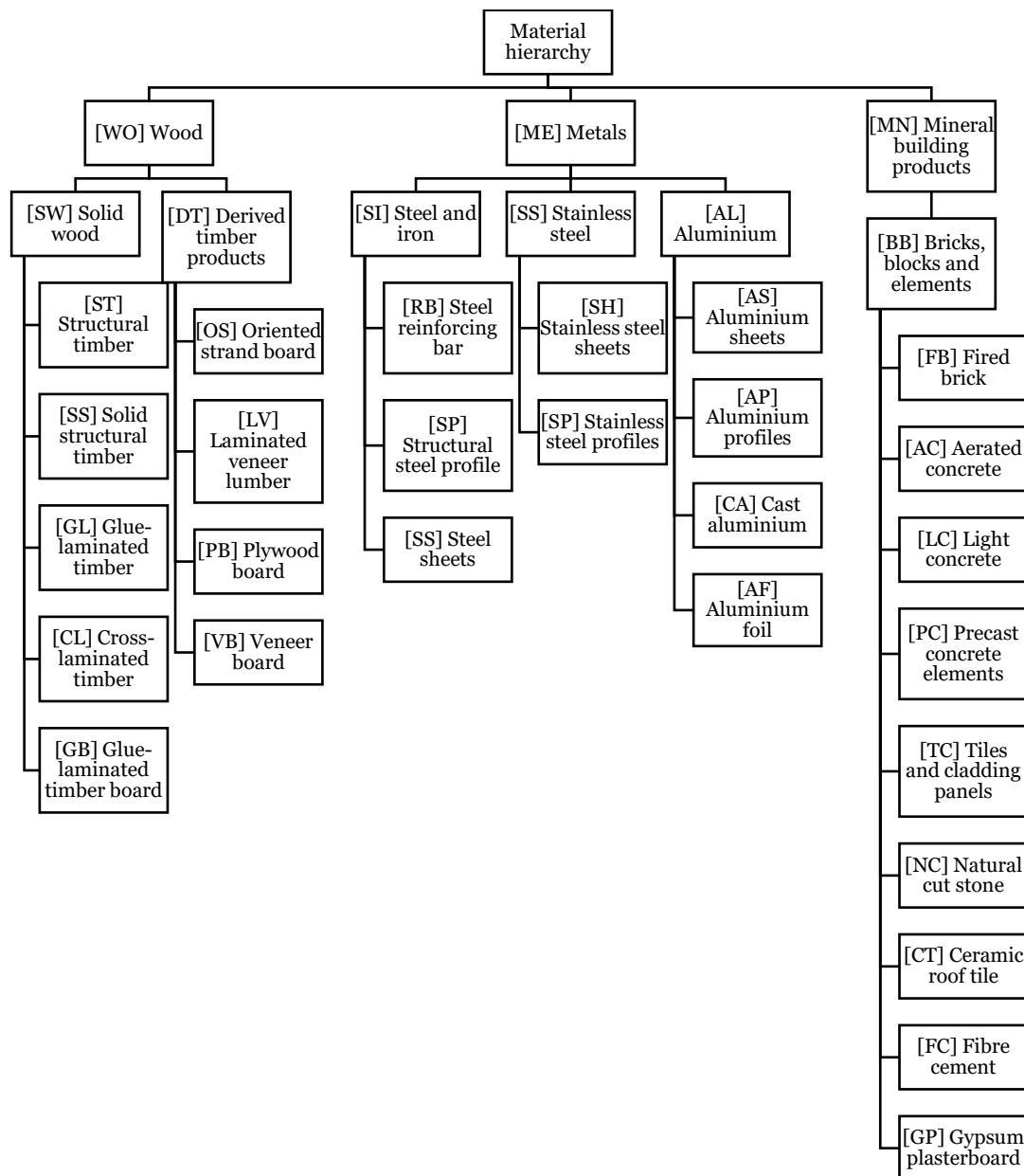


Figure 5. Material hierarchy of tags.

The two letter code preceding each node name serves as a machine readability tool and a label for designers for quick navigation. The labels in combination form a parseable code, where each position is reserved for a label of particular hierarchy, and each hierarchy segment is delimited. This way the code is a composition of labels of fixed length: [Spatial hierarchy labels set 1]_[Spatial hierarchy labels set 2]-[Functional hierarchy labels set 1]_[Functional hierarchy labels set 2]-[Material hierarchy labels set 1]_[Material hierarchy labels set 2]. As each item can have an unlimited number of tags assigned and the classification system must be expandable, this code organization allows for multiple sets of tags within each main hierarchy position delimited by an underscore “_”, and each hierarchy position is delimited by a dash “-“. Because each tag can be optional, the unused position in the code is marked with a placeholder “XX”. An arbitrary example of a code can be “RORA-XX-MEALAS_MEALAP” that would point to “Roof accessories” under “Roofs” with no tag selected for function and both “Aluminium sheets” and “Aluminium profiles” tagged as material. The code system should allow for unrestricted expandability, human readability, easy parseability for automation, however it must be noted that upon structural changes to a hierarchy a backwards compatibility dictionary of deprecated labels must be upkeep.

For the purpose of providing designer utility in the form of filtering by technical properties, a parallel set of performance attributes is linked to a corresponding tag sequence. Being modular and additive in nature, the format of linking a separate table of attributes to sequences of tags of various depth allows for easier management. The attributes include such parameters as load-bearing capacity, fire rating, acoustic insulation values, etc.

The hub functionality is realized via data tables representing product and element codes from existing classification systems. Each of these codes has a key in the mapping table which values are listing all matching tag sequence codes from the prototype schema. Due to the bidirectional nature of the mapping, more potential is unlocked for searching products and elements, and switching between classification systems within a project.

Thus, the metadata schema includes three parallel hierarchies of tags, a performance attribute set, mapping tables.

4.1.4 Interviews and evaluation

This subsection reports on the semi structured interviews conducted during the first cycle of the action research. The purpose of the interviews was to evaluate the prototype of the classification system, gather practitioner insights across multiple disciplines, and identify specific areas for improvement.

The interview protocol and the full interviewer notes are listed in Appendix 1. Below a high-level summary of the interviews and the most relevant points from participants' answers are presented.

When asked about classification system use in participants' workflows, most practitioners answered that they relied on at least one formal system, Talo 2000 was predominant, with secondary use of Tekla's built-in schema, IFC classifications, or manufacturer catalogues, however a notable minority worked without any formal taxonomy, depending instead on ad-hoc naming or brand directories.

A clear common theme emerged during the interviews about product discovery being held back. When asked about challenges in locating and specifying products or building elements, participants uniformly cited the absence of embedded data within three dimensional models. Several engineers explained that they often receive models containing only geometric solids with minimal metadata, which is usually limited to spatial hierarchy. Thus, models have no attached material, performance, or brand information, and engineers are forced into time consuming exchanges of spreadsheets and CAD drawings. One structural engineer observed that BIM frequently stops at design handover, and the data doesn't carry through to site, while another BIM manager noted that without a seamless link between Revit families and classification codes, we lose vital attributes in later stages. When asked which attributes matter most, technical performance (load capacity, fire rating, etc.) and material properties consistently ranked highest, with brand and cost following after the performance criteria were met. Sustainability metrics were increasingly referenced but remained secondary. Interoperability was universally deemed critical, though opinions varied on its current status. More senior practitioners emphasized the need for classification codes to persist seamlessly from initial concept through permitting, construction, and operation, while some MEP designers downplayed these issues, noting that they routinely supplement BIM data with Excel tracking. Nevertheless, there appears to be a consensus that a classification system that cannot be integrated into designing tools is of limited use.

These insights confirm that any effective classification system must address three core requirements: 1) integrated metadata within BIM deliverables; 2) flexible attribute filtering to surface performance and material data on demand; and 3) end-to-end code persistence for seamless data handover across tools and project phases.

In the second part of each interview, participants reviewed the prototype version of classification with the several parallel hierarchies. Feedback was gathered around five main themes:

1. Hierarchy and navigation: interviewees recommended adding numeric top level codes based on colloquially used systems for recognition, quicker navigation and filtering. Flexible inter-hierarchical navigation was suggested with a focus on user experience. Navigation and narrowing down through performance characteristics of items was suggested. Changing the spatial hierarchy to start from substructure and continue upwards was suggested.
2. Terminology and definitions: several practitioners noticed a lack of clarity in some naming choices. More intuitive naming for top levels of hierarchies was suggested with additional definitions for clarity.
3. Discipline perspectives: Structural and MEP practitioners suggested splitting or filtering the hierarchies based on discipline (e.g. architectural, structural, MEP). An example of such change could result in an element like “railing” to appear simultaneously under accessory and enclosure based on the specific designer. For architects, enclosure could be better suited, while for structural designers it would be better seen as an accessory. An additional split into load bearing and non-load bearing functional categories was suggested to be of use for structural engineers.
4. Scope extensions: HVAC engineers have noted shortcomings in the coverage and lack of deeper subcategorization. Other interviewees noted a need in more precise performance distinctions (e.g. sound insulation must be split into impact, airborne, etc.). CO2 footprint integration was suggested. It was also suggested by one interviewee to keep the system expandable.
5. Localization and mapping: for better adoption interviewees recommended embedding variations of tag names for various languages based on locally used systems, i.e. basing the translated names on widely accepted local terminology. Mapping to other systems was underscored. Short numerical codes were suggested as a machine readability enhancement and human readable labels were noted as an intuitive solution.

The feedback provides a framework for improving the prototype in Cycle 2, where the structure of hierarchies will be reviewed in addition to other aspects to reflect the feedback.

4.1.5 Reflection

Cycle 1 successfully transformed the conceptual goals into a tangible prototype and received direction for further improvement. A flexible, tag-based

framework was developed that includes spatial, functional, and material hierarchies, way to link performance attributes, and bidirectional mappings to existing systems. Feedback was received from nine practitioners across architecture, structural, MEP, and BIM coordination disciplines. The interviewees found the hierarchy structure logical and potentially useful, although they also showed areas for improvement in the prototype. These practitioner recommendations will directly inform Cycle 2 plan and action.

4.2 Cycle 2

4.2.1 Synthesis and plan adjustment

The evaluation in Cycle 1 outlined priority areas for enhancement in hierarchy structure, terminology, discipline specificity and user experience. The scope of the system was prompted to be increased to include translations for international utility, sustainability metrics, and other. The considerations are reflected in the following subsection.

Cycle 2 includes refinement and implementation of changes, where concerns from the evaluation in Cycle 1 are addressed. In addition, feedback and evaluation is carried out, however in Cycle 2, after refining the prototype, feedback was collected through an e-mail-based review. The section ends with a reflection on progress so far.

4.2.2 Refinement and implementation of changes

The second version of the classification system incorporates two types of changes: those driven directly by practitioner feedback and a second group of general refinements that improve the overall clarity. However, since both types of changes are closely related and sometimes go together, they have been described uniformly.

A potential issue was recognized in confusion between “spatial” and “building space” therefore to make the focus on physical elements rather than building spaces, the “spatial” hierarchy was renamed to “element” hierarchy. This change more clearly indicates that the top level categories represent building elements (e.g., foundations, superstructure, walls, roofs) rather than spaces. In version 1, the roof categories appeared first, which some practitioners found counterintuitive when navigating from ground level upward. Version 2 reorders the element hierarchy so that “Substructure” appears at the first level, followed by “Superstructure”, “Floors”, “Walls and barriers”, and “Roofs”. Although numeric top level codes were proposed, the letter-based labels were retained (e.g. “SU” for Substructure, “ST” for

Superstructure) because they already provide both unhindered machine-readable parsing and human-readable labels.

Interviewees noted that some labels were unclear, in combination with that the focus was set for the element and functional hierarchies to answer distinct questions, where the element hierarchy would answer the question “Where?” and the function hierarchy would answer the question “What for?” or “What function?”. Therefore, tags in both hierarchies were revised to more clearly answer these questions, and naming was changed in several places for more clarity. This includes splitting the “Insulation sound” tag into “Insulation sound – airborne” and “Insulation sound – impact”. An example of answering the questions: “Where?” – “Beams and joists”; “What?” – “Anchor”. By making each tag more intuitive, users can skim a list and immediately understand whether they are selecting a component’s location (Element) or its role (Function). Structural and MEP interviewees suggested that some elements (for example, “railing”) might be better classified differently depending on whether an architect or a structural engineer is using the system.

To accommodate these discipline driven viewpoints without duplicating the entire hierarchies, the second version introduces broader level 1 Function categories: Structural, Envelope and others. The new level 1 should direct the user towards their desired items, however due to the precision obtained with the parallel tagging system, any sort of filtering is available in a user interface implementation. At the same time, this solution solves the issue of separating load-bearing items from non-load-bearing by making them implicit. The MEP coverage was slightly expanded as a proof of concept, with the focus on functional tags. In version 2 one of the expanded MEP-centric function tags include “Ventilation” with subcategories “Fume extraction”, “General space ventilation” and “Smoke extraction.” The updated tree structures of the “element” and the “function” hierarchies are shown in Figure 6 and Figure 7 respectively, while the “material” tree structure stayed unchanged as seen in Figure 5.

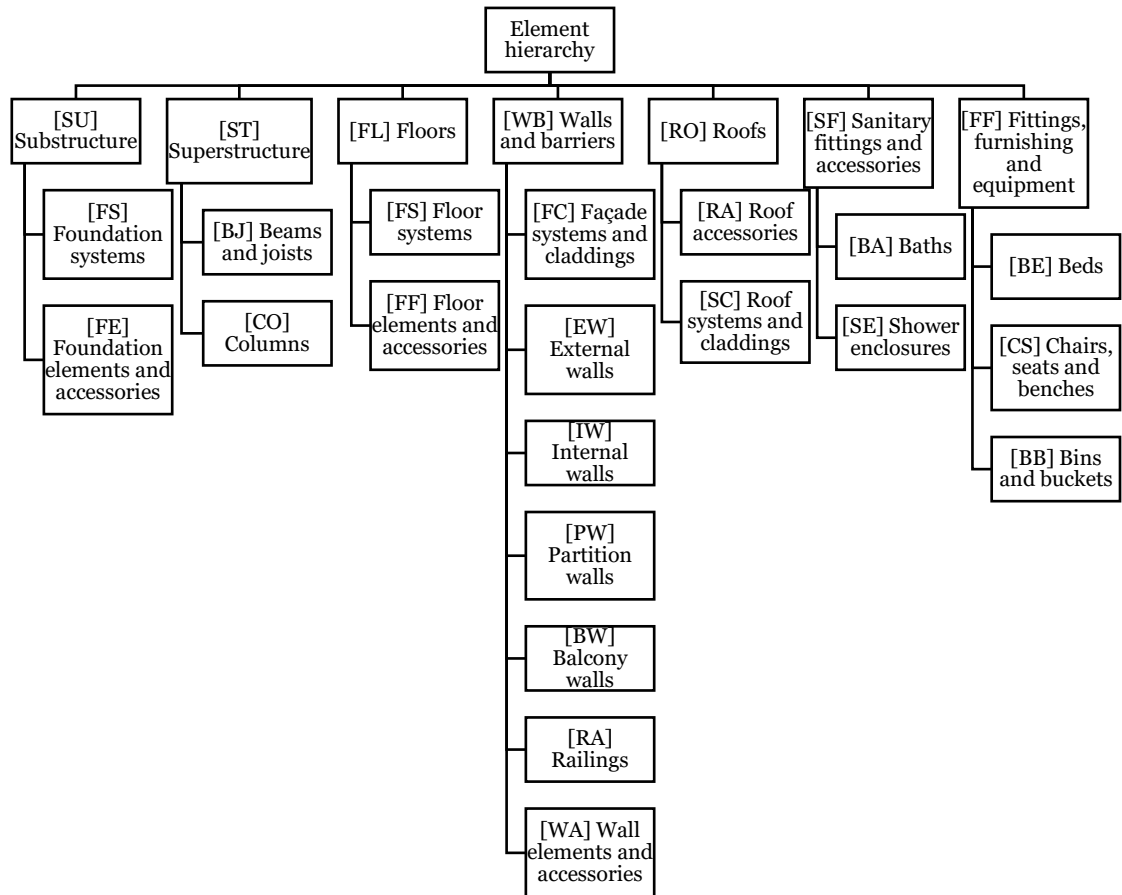


Figure 6. Element hierarchy of tags.

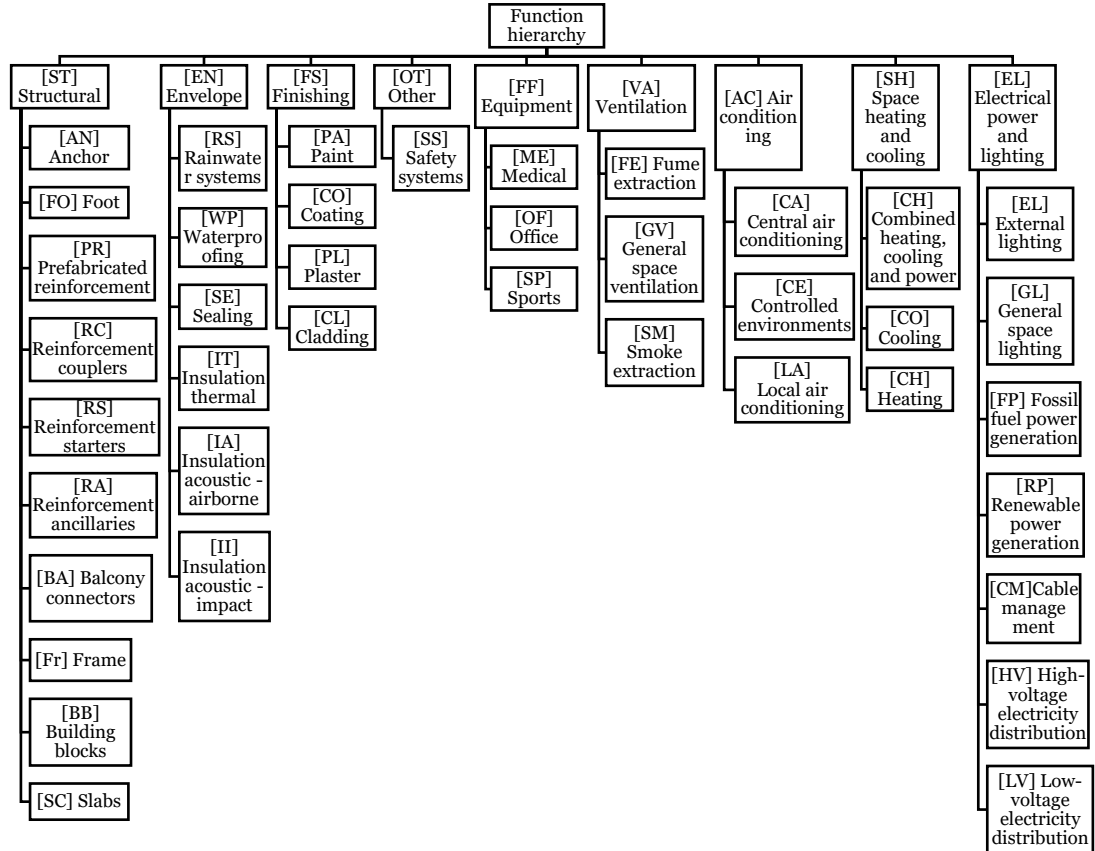


Figure 7. Function hierarchy of tags.

Extensibility is intrinsic to this prototype: because tag sequences are delimited within a fixed length code, new subtags can be appended or entire axes expanded without breaking existing entries. Thus, new requirements, such as new material types, functional roles, or system categorizations can be incorporated. In cases of structural changes backward compatibility can be preserved via a dictionary of deprecated labels.

Addressing concerns about discipline specific orientation and filtering based on aspects such as “load bearing” or “non-load bearing”, it was decided that these are derivable characteristics that do not have to be embedded in the prototype concept, as the desired functionality of filtering such objects can be implemented separately. What is more, element and functional hierarchy should already be implicitly reflecting these aspects, leaving space for further user experience enhancements via practical implementations of the system.

Additionally, in response to practitioners’ emphasis on environmental considerations, sustainability metrics are integrated alongside performance characteristics as global, universal data fields.

A separate translation table was also established to manage multilingual and regional variations in tag terminology. Each tag name has an alias term that is based on terminology used in national systems or a widely accepted term.

After the changes to the prototype, the metadata schema includes three parallel hierarchies of tags, unified set of attributes such as performance characteristics and sustainability metrics, translation table, mapping tables.

4.2.3 Feedback and evaluation

The objective of the second feedback round was to evaluate the changes implemented to the prototype in Cycle 2. An e-mail-based review was conducted in which the participants were presented with the new structure of the system with the description of changes as seen in the previous subsection. Out of nine original interviewees only three responded to the email, however with representation from architecture, structural engineering, and MEP fields. The details of the second round of feedback through e-mail including prompts for practitioners and notes from the responses can be found in Appendix 2.

All respondents noted that the new version is more intuitive and includes necessary aspects for potential use. The practitioner with architectural background noted that an inclusion of extensions in the form of “special” categories can be beneficial, such as “Special wall” or “Special ventilation equipment”. According to them, these additional categories could be useful for the system adoption by designers in demanding projects, by leaving a customizable layer. The MEP engineer recommended aligning tag names with industry conventions more thoroughly and including discipline prefixes for clearer scope and distinction. The structural engineer expressed

satisfaction with the clarity of the system, however emphasized the importance of item filtering by parameters, such as, for example, fire rating.

4.2.4 Reflection

Cycle 2 resulted in the implementation of changes and improvements based on practitioner insights and feedback. The prototype's taxonomy was refined and additional functionality was added to the concept, such as translations and sustainability metrics. The hierarchy structure was expanded to include more categories, in addition to a clearer structuring.

The changes were communicated to the interviewees for a second round of feedback through email. Although only three practitioners have responded, a good representation of disciplines was preserved (architecture, MEP, structural). The second version of the prototype received good reception and constructive comments on improvements. The gathered feedback guides the final adjustments in Cycle 3, where the prototype is followed up with the suggested changes, the schema is formalized, the implementation is documented, and the result is evaluated.

4.3 Cycle 3

4.3.1 Synthesis and plan adjustment

Cycle 2 incorporated changes and evaluated the prototype's second version. Based on the feedback in Cycle 2, the prototype concept mostly reached an appropriate state, with some minor changes still required.

Cycle 3 includes refinements in accordance with feedback from Cycle 2, however, it is additionally shifting from conceptual development to outlining the practical implementation. The complete taxonomy and concept with additional functionality is represented in its final form and the metadata schema is formalized with mapping tables and tables of performance characteristics. Lastly, a final practitioner evaluation of the latest prototype is carried out through an e-mail-based review.

4.3.2 Final refinements and schema formalization

The third version of the classification system prototype includes changes based on feedback collected in Cycle 2. It includes additional user-defined "special" categories that can be used to expand the classification system for atypical entries, which allows for their separate sorting. It also adds optional discipline tags used for filtering by architectural, structural, and MEP

disciplines as seen in Figure 8. Figure 9, Figure 10, and Figure 11 depict updated Element, Function, and Material hierarchies respectively.

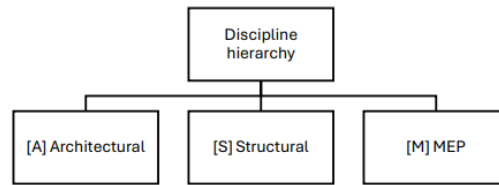


Figure 8. Discipline tags.

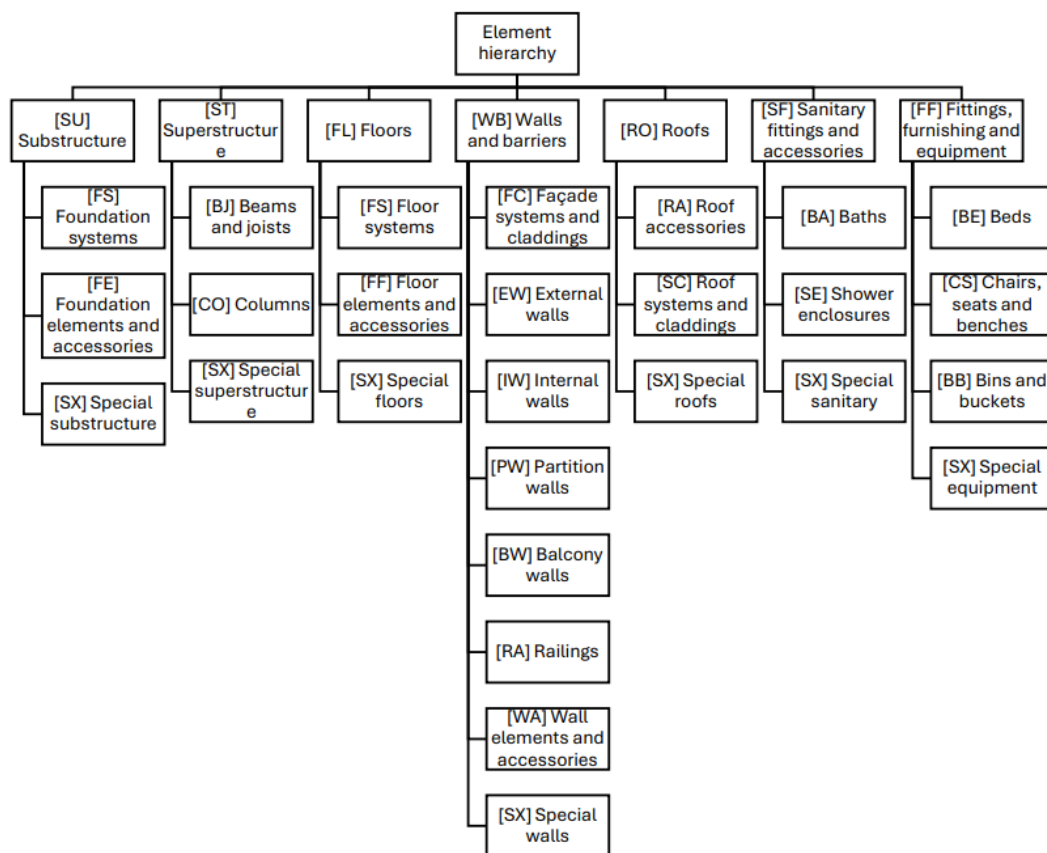


Figure 9. Element hierarchy of tags.

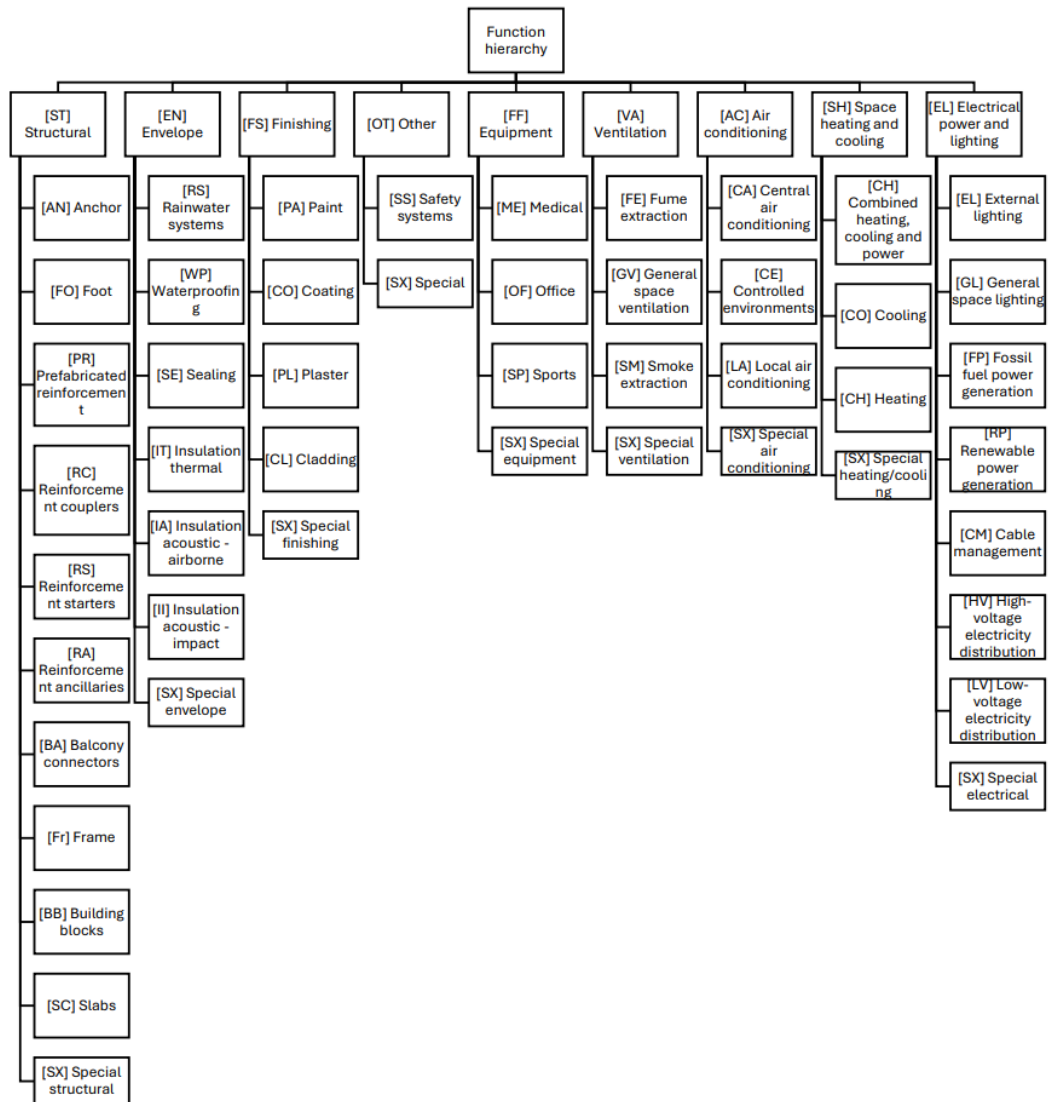


Figure 10. Function hierarchy of tags.

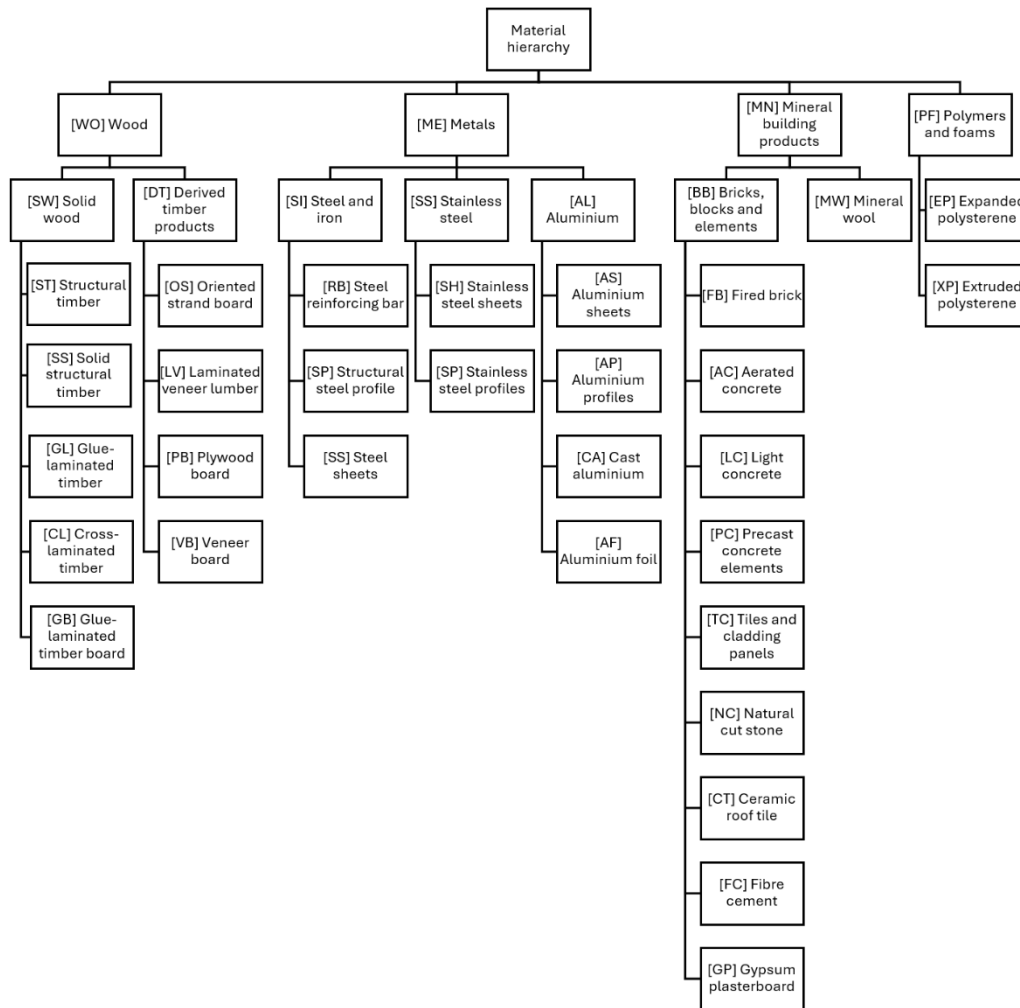


Figure 11. Material hierarchy of tags.

An arbitrary example of a code can be “A_S-RORA-XX-MEALAS_MEALAP” that would point to “Roof accessories” under “Roofs” with no tag selected for function and both “Aluminium sheets” and “Aluminium profiles” tagged as material. “A” and “S” in the beginning point to the item being relevant for both architectural and structural disciplines. It is important to note that current scope of the classification system is not comprehensive and serves as a step towards expansion with new tags added to each hierarchy.

To support parameter-based filtering, product search and comparison, as well as smooth information flow, the prototype uses an attribute schema that links tag sequences to standardized properties. Table 2 serves as a depiction of the principle of linking a classification code (or a tag sequence) to attributes, which can be performance characteristics and other parameters.

Table 2. Attribute mapping.

Attributes				
Name	Unit	Values range	Code	Tag sequence
Width	m		XX	Any
Length	m		XX	Any
Global Warming Potential (GWP100), A1-A3, EPD Total	kg CO ₂ -eq		XX	Any
Reaction to fire		A1...F	XX	Any
Wind load resistance	kN/m ²		WBFC	Walls and barriers - Façade systems and claddings
Thermal resistance	m ² ·K/W		ENIT, WBFC, WBEW, WBBW, ROSC, FLFS	Envelope - Insulation thermal, Walls and barriers - Façade systems and claddings, Walls and barriers - External walls, Walls and barriers - Balcony walls, Roofs - Roof systems and claddings, Floors - Floor systems

Attributes can be generic geometric descriptors, sustainability metrics, and performance characteristics, as well as other values suitable for the format. The attributes and their values may come from a variety of sources, from generic information, through manufacturer provided information, to publicly available sources with APIs. Each attribute can also be a global (universal) attribute, which means it is applied to all classification tags or any sequence of tags. Geometric descriptors are represented by fields such as “Width” or “Length”, both they and the sustainability metrics are global and universal attributes in this version. Performance characteristics such as “Wind load resistance” and “Thermal resistance” follow European naming and unit conventions and are linked to the relevant tag sequences based on normative sources. In practice, this linkage is supported by the SKOL product specification guidelines, which indicate characteristic properties by product type.

The principle of mapping to other classification systems, and, therefore, serving as a medium for cross-classification mapping is reflected in Table 3 and Table 4. The mapping in the prototype classification system is not done in a strict match manner, but for any matches in a code sequence. This means that if the target prototype code sequence (like “AS-RORA-XX-MEALAS_MEALAP”) includes the mapped prototype code “RORA” it will point toward the equivalent category in the Talo 2000 classification. In cases where there is several codes with a coma (“ROSC, ENIT”) it means that the target sequence must include both these codes.

Table 3. Talo 2000 project classification mapping table.

Talo 2000		Prototype	
Code	Name	Codes	Tag sequences
1212	Enclosure walls, foundation columns, foundation beams	SUFS	Substructure - Foundation systems
1211	Footings	SUFE	Substructure - Foundation elements and accessories
1213	Special foundations	SUSX	Substructure - Special substructure
1241	External walls	WBEW	Walls and barriers - external walls
		WBFC	Walls and barriers - Façade systems and claddings
1262	Roof substructures	ROSC, ENWP	Roofs - Roof systems and claddings, Envelope - Waterproofing
		ROSC, ENSE	Roofs - Roof systems and claddings, Envelope - Sealing
		ROSC, ENIT	Roofs - Roof systems and claddings, Envelope - Insulation thermal
		RORA	Roofs - Roof accessories
1263	Roofings	ROSC	Roofs - Roof systems and claddings

Table 4. Uniclass classification mapping table.

Uniclass		Prototype	
Code	Name	Codes	Tag sequences
EF_20_05_30	Foundations	SUFS	Substructure - Foundation systems
EF_25_10_25	External walls	WBEW	Walls and barriers - external walls
Pr_25_57_06_53	Mineral wool insulation	ENIT, INMW	Envelope - Insulation thermal, Insulation materials - Mineral wool
Pr_20_29_03_86	Structural anchors	STAN	Structural - Anchor

Additionally, separate classification systems can be cross-linked via the shared prototype tag sequences when both external codes correspond to the same prototype tag. For example, External walls in Talo 2000 and in Uniclass are linked through the prototype tag “WBEW” (Walls and barriers – External walls). This linkage increases transparency by exposing both overlaps and gaps between systems, and it enables querying for the closest available counterpart where exact one-to-one correspondences are not present.

The translation or alias table is part of the finalized schema but is not reproduced here to keep the section focused and because terminology is region- and language-specific. In the model, each tag maps to a table of names and other regional aliases. Figure 12 depicts the model of the prototype including its constituents and their interrelations.

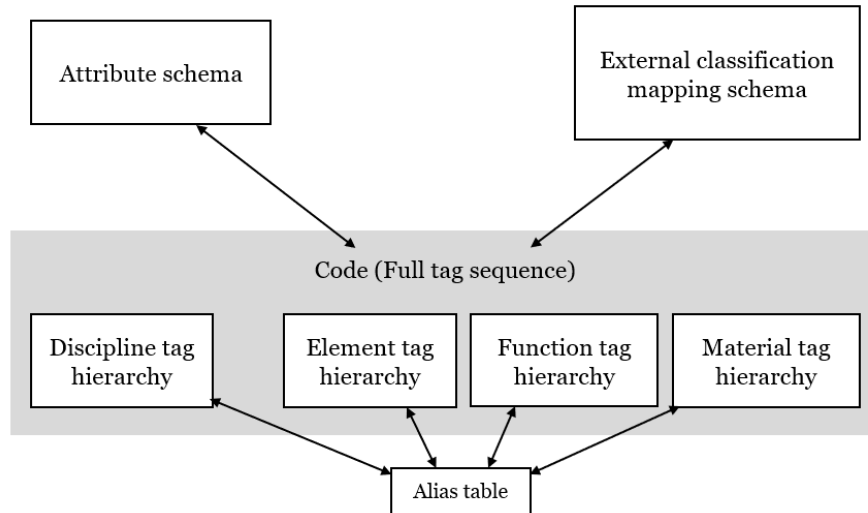


Figure 12. Classification system prototype model.

4.3.3 Feedback and evaluation

The final e-mail review asked participants to assess the third and final version of the classification system prototype by evaluating the implemented changes and by mentally stepping through typical tasks such as locating products and elements, filtering by attributes, cross-referencing with Talo 2000 classification system. The respondents were the same three practitioners who responded in the previous round, keeping the representation of architectural, structural, and MEP backgrounds. The details of the third round of feedback including practitioner prompts and response notes are provided in Appendix 3.

Overall, the feedback was positive with practitioners acknowledging the utility of the prototype classification system and approving of the schema and the implementation concept. The structural engineer found the schema intuitive and approved the logic for mapping attributes and external classification systems. A point was made by the MEP engineer that their fields' coverage is still lacking and needs expansion. Both the MEP engineer and the architect were approving of the overall results, however they had minor reservations about the code format being initially confusing for users, although functional.

4.3.4 Reflection

Cycle 3 included required refinements defined by feedback in Cycle 2 and consolidated the prototype into a coherent and usable schema, while confirming the utility of the tag-based approach, centered around designers. The refinements included minor additions to the tag hierarchies in the form of

“special” categories for atypical items. The latest tag hierarchy, code logic, attribute mapping, and external classification mapping logic was formalized with examples.

The full concept with practical implementation layout was communicated to the practitioners for review and evaluation. Feedback in the last round was consistently positive across disciplines, showing that the concept and its implementation logic is understood and can support designers’ tasks. Nevertheless, one respondent noted the shortcoming of the underdeveloped state of the classification breadth, which is in line with the scope of the development performed in this thesis. The final classification system prototype version remains incomplete with current coverage and implementation outlining being illustrative rather than exhaustive.

Overall, Cycle 3 achieved a result that shows a clear promise in a structured, interoperable, multi-hierarchical classification system that can be expanded and developed further without breaking earlier decisions and can serve as a translation layer between other classification systems. The received feedback showed the following steps for immediate improvement: extend the scope of the system by aiming to achieve exhaustive coverage, continue linking attributes to tag sequences using normative sources such as General product specification guidelines for structural types by SKOL ry, continue mapping external classification systems to tag sequences, develop a simple alias table for international translations, provide user guidance for code interpretation. The evaluation in Cycle 3 relied on document-based walkthroughs with a small group of practitioners, which is appropriate for this stage but does not replace task-based testing. The testing should include a pilot prototype UI and selected BIM, quantity, and LCA workflows to validate performance in real tasks. These actions would convert the prototype into a robust and adoptable asset for design practice.

5 Discussion

The research including literature review, comparative analysis, and the AR cycles resulted in the research questions being answered. The analysis of current classification systems in Europe and other key regions created an overview of what systems are used and what problems they are solving, answering the first research question. The reviewed classifications are different in foundational logic, scope, and depth, while not simultaneously providing broad coverage, day-to-day usability for designers, and straightforward interoperability.

Research question two was answered during the development of the classification system prototype, showing that a tag sequence approach can act as a bridge between unique systems. Two separate systems often can have significant mismatches, even though they are created on the basis of the same normative literature, which is bound to leave gaps preventing straight mapping. Therefore a tag sequence approach is beneficial due to creating flexible connections, allowing for mapping a tighter categorization to a higher order one between two separate systems, in addition to highlighting the gaps.

For the third research question, knowledge gained from the literature review together with practitioner input acquired in the AR revealed the needs for embedded and navigable structure for BIM deliverables, filtering by performance characteristics and other attributes, persistence of identifiers across tools and phases, discipline perspectives.

Research question four raised the topic of linking classification categories to attributes, which was developed in the prototype. Linked attributes are represented by performance characteristics, sustainability metrics, geometric dimensions, and other. This functionality was acknowledged by the practitioners as a useful tool for product discovery and comparison. A practical implication is that the attribute layer can be populated and maintained using existing product data conventions. In particular, the schema can host attribute sets sourced from established models such as ETIM or nationally used product data standards.

For research question five, the combination of parallel “element”, “function”, and “material” tags, optional discipline tags, and explicit mappings supports both flexible product discovery and structured data exchange. The compact delimited tag codes and attribute tables are designed to be machine-readable, enabling querying and creating a basis for automated search and selection. Lastly, the AR process itself proved effective and serves as an answer to the inquiry of practitioner-empowered incremental development. Successive rounds of feedback led to concrete changes and

refinements, leading to an overall recognition and approval of the prototype by the practitioners.

The results of this thesis appear to fit within and build on top of the existing body of knowledge. The prototype's three parallel hierarchies are drawn from the structuring principles of ISO 81346-12. The prototype does not aim to replace or simply add one more classification system to an already extended list of them, however it is adding on top and allowing the existing classification systems to be utilized in new ways and with less friction in international projects. The most consequential contribution is the multilateral, tag-based approach that treats classifications as additive tag sequences, rather than a fixed tree. This allows both flexible navigation and precise classification, while also allowing the system to serve as a mapping hub between external systems. The hub functionality, in addition, can be served not only as a cross-referencing tool, but due to the schema of the prototype it allows any external classification system to be used with attribute linking. Due to the additive nature of the classification and the attribute schema allowing value ranges to be specified, the model is agnostic to whether the products are make-to-stock or engineer-to-order. The functionality of the model, though designed from the perspective of designers, has potential to provide benefits in contractor-led product selection too. The outcome is intentionally illustrative rather than operational, its purpose is to demonstrate feasibility and utility, that directly serves as a tool for more efficient BIM workflows.

Finally, practitioner input was essential in directing the work and occasionally unexpected. For example, several participants emphasized the role of a brand in their product selection habits over raw performance or other criteria. Yet, this observation supports the proposed model, as with consistently tagged items and assigned attributes, filtering and comparing by attributes creates a pathway to discover equivalent or superior alternatives in unfamiliar brands. Taken together, the results suggest that the combination of the attribute layer anchored on standards and the tag sequences is a potential route towards semi-automated or automated product search, while preserving human readability.

6 Conclusions

This thesis examined existing classification systems from the perspective of designers and developed and evaluated a harmonized, designer-oriented classification system prototype that simplifies the organization of construction products and elements, and supports designer workflows. The full prototype model including structure, tag hierarchies, code format, attribute and classification mapping logic is provided in Appendix 5. The main contribution is an illustrative classification system prototype supporting product discovery, precise classification, and that serves as an augmentation for existing classification systems and a bridge between them. The proposed tag sequence structure is readable and parseable in practice, supports linking attributes for filtering by performance characteristics and other, enables transparent and flexible cross-classification mapping. Practitioner feedback across two e-mail rounds confirmed that the concept is understandable and useful for typical tasks. Nevertheless, practical testing should be performed with a pilot prototype UI and selected designer workflows to validate performance in real tasks.

For practitioners, the implication is to use the prototype classification system model as a bridge rather than a replacement for existing classification systems. Embedding the tag sequences representing the three parallel hierarchies of tags in BIM objects and product libraries, attaching and enriching the attribute sets can improve product discovery, comparison, and documentation without disrupting established deliverables. Maintaining the mapping tables to existing classification systems allows for their use as the main classifications together with the prototype functionality, and allows for more seamless classification in international projects. Simple alias tables can also support local terminology, and UI can help mitigate unfamiliarity with the prototype's codes.

The study has limitations. In particular, the practitioner sample is small and the development ended with only three active participants. Other limitations include limited coverage of the prototype system, final evaluation relying on e-mail reviews, in addition to the attribute and external classification system mapping being illustrative.

Further research should extend the taxonomy while preserving backward compatibility, especially if the prototype proceeds to practical implementation. The next step is to validate the model in real design tasks, testing it across different product types and, where relevant, across workflow practices, such as designer-led and contractor-led. An evaluation should be carried out to measure how far the prototype model structure enables automated product search and selection, therefore reducing manual

interpretation. In parallel, external classification mappings, the development and normalization of the attribute set, and the alias table should be completed using official classifications and normative literature. The schema is intentionally designed to evolve without breaking earlier decisions. Defining clear governance and update processes will help sustain continuous improvement.

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Appendices

Appendix 1. Interview protocol and notes – Cycle 1

Interview Format and Purpose

The interviews were conducted in a semi-structured format. This format was chosen to accommodate the diverse professional backgrounds of the interviewees, primarily designers and engineers, and to allow for variation in the conversations depending on the interviewee's expertise. The intention was not only to gather specific data through targeted questions, but also to encourage open discussion and allow participants to share additional insights. This approach was chosen to capture a broader understanding of classification system use in practice and to identify current challenges and needs in the field.

The interviews were structured in two parts. The first part included general questions about the interviewee's background, current classification system usage and related challenges. The second part focused on gathering feedback on the prototype classification system.

The questions listed below served as a guideline and were used flexibly to enable follow up questions or further discussion.

Part 1:

1. Could you describe the nature of your work and how many years of professional experience you have?
2. What phases of a construction project do you typically work on or are involved in?
3. Do you currently use any classification systems in your workflow? If yes, which ones?
4. What are the biggest challenges you face when trying to find, organize, or specify products or building elements?
5. Which product or building element attributes are most important to you in your work? (e.g., technical performance, material, cost, brand, sustainability)
6. How important are interoperability and information flow between systems or tools in your daily work?

Part 2:

In the second part of the interview participants were shown a draft version of the prototype classification system developed for a limited group of construction elements and products. In some cases the prototype was sent in advance of the interview, in others presented during the interview. The aim was to gather feedback on the system's structure, clarity, potential usability, and compatibility with other classification systems and design tools. Participants were encouraged to provide suggestions for improvements, expansion and to share any additional observations based on their professional perspective.

Interview Notes

Interview 1

Role and Experience: Executive Director, 15 years of experience in structural engineering (new construction)

Project Phases Involved: All phases

Classification System Usage: Tekla Structures; references YTV 12

Challenges Identified:

- Difficulties in finding appropriate products.
- Materials and structural types often predefined.
- Noted disconnect between BIM and CAD (data vs. visual info).

Important Attributes/Product Discovery:

- Searches typically begin with material or basic technical performance, with a lot of back and forth between product and performance.
- Emphasized separating performance metrics for each constituent of an assembly.

Interoperability:

- Considered highly important, especially across project stages. Noted excessive back and forth flow of data in design.
- Noted that BIM data rarely persists into later project phases.

Additional Comments:

- Designers often resist transparency, which limits interoperability.
- BIM use frequently ends at the design stage.

Feedback on Prototype:

- Emphasized need for unified elements across systems. The same building element (e.g. a wall panel) should carry the same identifier or metadata regardless of classification system.
- Codes (or numerical indexes) and names are both important in classification schema nodes. Codes can be utilized to show broader categories.
- Numbering or indexing is useful for hierarchy (the interviewee meant navigating through hierarchical schema by using indexes as an anchoring point).

- The classification system draft is not very structured.
- Possibly more perspectives can be added to the classification system, meaning perspectives for architects, structural engineers, MEP etc.
- Suggested reorganizing hierarchy from substructure upwards.

Interview 2

Role and Experience: Structural Designer, 4.5 years

Project Phases Involved: All phases

Classification System Usage: Not specified

Challenges Identified:

- No systematic method for information transfer.

Important Attributes/Product Discovery:

- Usually no systematic way.
- Emphasis on manufacturer (e.g. trusted manufacturer), then performance characteristics.
- Revisions common due to manufacturer switches.

Interoperability:

- Interoperability considered essential.
- Lack of embedded data in 3D models mentioned. Noted that the 3D models they receive often are purely graphical.

Feedback on Prototype:

- Noted that Swedish systems are pretty well developed and take note from what has been done there.
- Classification should be deep with precise grouping.
- Functional point of view is important.
- Suggested possible hierarchies: element, material, purpose.
- Mentioned that designers should be able to see differences between elements, and items could be compared by characteristic parameter values.
- Suggested to split classification per designer (Architectural, structural, MEP).

Interview 3

Role and Experience: Structural Engineer (10 years), currently in construction industry development

Project Phases Involved: All phases

Classification System Usage: Talo 2000, IFC classifications, other (depending on customer)

Challenges Identified:

- Lack of synchronization between geometry and product data.

Important Attributes/Product Discovery:

- Typically begins with brand or manufacturer, then technical performance.

Interoperability:

- Highlighted the need for consistent naming and attribute sets.
- Highlighted the importance of IFC classes.

Additional Comments:

- Mentioned that material information and other material related information is important (e.g. CLT lamella number).

Feedback on Prototype:

- Considered different perspectives on one node, node like “Railings” can be viewed as an accessory (unlike their location under Walls and Barriers in the draft system).
- Recommended separation of structural and non-structural elements.
- Sound insulation can be split into impact, airborne and other types.
- CO₂ footprint could be linked to specific nodes.
- Codes should be short, memorable, machine readable, and embedded in BIM objects.
- Noted that mixed up items between architectural and structural disciplines are inevitable.
- Suggested to focus on locally used systems.
- Numbering can be useful for remembering (e.g. designers remember codes of Talo2000).

Interview 4

Role and Experience: Architectural BIM Content Creator/Architectural Designer, 10 years

Project Phases Involved: Design and BIM management

Classification System Usage: Talo 2000, YTV

Challenges Identified:

- Difficulties in finding the right category for an object.
- Design software like Revit may have different and not matching categories.
- BIM content is often not unified (not harmonized).

Important Attributes/Product Discovery:

- Product discovery starts with technical parameters and then brand.
- Sustainability, visual representation are also important.

Interoperability:

- Interoperability should be a part of the design flow.
- Noted that national standards must support interoperability.

Feedback on Prototype:

- Found the structure clear and logical.
- Suggested splitting hierarchies by discipline (architectural, structural, MEP).

Interview 5

Role and Experience: Structural Engineer, 5 years

Project Phases Involved: Consulting design and detailed design

Classification System Usage: Tekla Structures internal classification

Important Attributes/Product Discovery:

- Product selection typically begins with brand, then performance. Sometimes agreements with manufacturers dictate particular brand.
- Sustainability and cost also considered.

Interoperability:

- Highlighted the importance of interoperability and mentioned that information can be lost during different stages, resulting in mistakes on site.

Feedback on Prototype:

- Suggested manufacturer tagging and filtering.
- Proposed including performance data such as load capacity or fire rating.
- Recommended ability to filter load-bearing vs. non-load-bearing items.
- Recommended to separate structural elements from non-structural.

Interview 6

Role and Experience: Residential Architect and Urban Planner, 6 years

Project Phases Involved: All post-draft stages (excluding site supervision)

Classification System Usage: Talo 2000

Challenges Identified:

- Lack of international compatibility.

Important Attributes/Product Discovery:

- Start with basic parameters like dimensions, then sustainability metrics and materials and costs are determined with the client.

Interoperability:

- Lack of BIM usage in some contexts negatively affects workflow.

Feedback on Prototype:

- Suggested multilingual translations.
- Supported letter-based codes for clarity.
- Emphasized need for a simple but expandable system.
- Proposed robust tagging.
- Mentioned an idea of potential image-based classification.

Interview 7

Role and Experience: HVAC Designer and Project Manager, 11 years

Project Phases Involved: All phases

Classification System Usage: Talo 2000, LVI numbers

Important Attributes/Product Discovery:

- Brand selection mentioned as one of the most important attributes.
- All attributes considered important.

Interoperability:

- Considered less critical. Interoperability managed through BIM and Microsoft Excel workflows.

Feedback on Prototype:

- Concerned that engineering products are underrepresented in the draft version.
- Suggested stronger MEP representation.

Interview 8

Role and Experience: MEP/HVAC Engineer and Project Manager, 7 years

Project Phases Involved: From concept design to commissioning

Classification System Usage: MagiCAD with built-in product catalog

Challenges Identified:

- Information fragmentation and lack of standardization

Important Attributes/Product Discovery:

- Technical performance mentioned as main attribute, followed by material and brand.

Interoperability:

- Considered important, however noted a lack of information and data fragmentation. Lack of standardization interdisciplinary workflows.

Feedback on Prototype:

- Suggested that naming should be standardized.
- Tables must be intuitive and recognizable.
- Suggested a more general schema for expandability.
- Recommended further expansion of MEP/HVAC categories.

Interview 9

Role and Experience: Structural Engineer, 14 years

Project Phases Involved: Primarily early-stage design; all phases

Classification System Usage: None currently; experience with BEC

Challenges Identified:

- Underutilization and lack of standardization of data.

Important Attributes/Product Discovery:

- Typically begins with material, then considers performance.

Interoperability:

- Sees room for improvement in workflows.

Feedback on Prototype:

- Proposed classification split by discipline (architectural, structural, MEP, fire engineering).
- Expressed the need for a functional-based hierarchy and recommended refining tables to reflect clearer logic.

Appendix 2. E-mail review prompts and notes – Cycle 2

Format and Purpose

The second round of feedback gathered practitioner input on the state of the second version of the classification system prototype via informal e-mail exchanges. Participants received annotated screenshots of the hierarchies describing implemented changes. In addition, a set of open prompts was put forward. Responses arrived as freeform e-mails. Key points were extracted into notes to maintain consistency with the interview notes in Appendix 1. Participants' names remain anonymous.

Together with annotated hierarchies, the following prompts were included:

1. What works and what doesn't in the current structure?
2. Is something missing from the concept?
3. Are tags and their codes clear and consistent?
4. Does the structure seem easy to grasp?
5. What could be improved?
6. Consider h

Notes

Response 1

Role and Experience: Structural Engineer, 5 years

What works:

- Clear and logical structure.
- Noted that tags are understandable, however emphasized consistency across similar categories.

Suggestions:

- Emphasized importance of filtering by performance characteristics.

Response 2

Role and Experience: MEP/HVAC Engineer and Project Manager, 7 years

What works:

- Intuitiveness improved compared to the previous version.

Suggestions:

- Tag names should be aligned with industry conventions.
- Discipline prefixes should be added for the codes to make the scope clearer and help more easily navigate.

Response 3

Role and Experience: Residential Architect and Urban Planner, 6 years

What works:

- Noted improvements in clarity overall.

Suggestions:

- Add special categories that could be utilized in demanding projects where some objects are too unique.

Appendix 3. E-mail review prompts and notes – Cycle 3

Format and Purpose

The final round of feedback gathered input from practitioners on the third (final) version of the classification system prototype via informal e-mail exchanges. Participants received a document containing the full structure of the classification system, explanation of changes, and examples of functionality implementation with explanations (similar to section 4.3.2). Together with the document the participants were prompted to assess and evaluate the prototype by mentally stepping through typical tasks such as locating a product or an element, filtering by attributes, and cross-referencing with Talo 2000 classification system. Responses arrived as freeform e-mails. Key points were extracted into notes.

Notes

Response 1

Role and Experience: Structural Engineer, 5 years

General:

- The schema is intuitive and the included functionality is useful.
- Logic in mapping performance characteristics and other classification systems makes sense.
- Noted that performance characteristics inclusion is a big task and is ever-ongoing.

Response 2

Role and Experience: MEP/HVAC Engineer and Project Manager, 7 years

General:

- Expressed the strong need to extend the coverage further, however is approving the results.
- The code system seems functional, but may be a bit confusing from the start.

Response 3

Role and Experience: Residential Architect and Urban Planner, 6 years

General:

- Approved the performed work and the system structure.
- Expressed a slight confusion with the code system, however noted that it is to be expected.

Appendix 4. Table drawn from SKOL ry General product specification guidelines for structural types

Finnish	English translation	Standard
90.011 Perusmateriaalit	90.011 Basic materials	
sementti	cement	EN 197-1
muuraussementti	masonry cement	EN 413-1
muurauslaasti	mortar	EN 998-2, SFS 7001
betonikiviaines	concrete aggregate	EN 12620, SFS 7003
laastikiviaines	mortar ingredient	EN 13139, SFS 7040
kiviaines maa- ja vesirakentamis- essa	aggregate in land and water construction	EN 13242, SFS 7005
luonnonkivilaatat verhoiluun	natural stone tiles for cladding	EN 1469, SFS 7019
luonnonkivilaatat	natural stone tiles	EN 12057, SFS 7019
luonnonkivi lattioihin ja portaisiin	natural stone for floors and stairs	EN 12058, SFS 7019
Betoniteräket	Concrete steels	
hitsattavat raudoitustangot ja raudoitusverkot	Weldable rebars and rebar meshes	SFS 1300
Betoniteräket. Kylmämuokattu ruostumaton betoniteräs B600XA, B600XB ja B600XC sekä ruostumattomat betoniteräsverkot.	Concrete steels. Cold-formed stainless concrete steel B600XA, B600XB and B600XC and stainless steel concrete meshes.	SFS 1259
Raudoitteet	Reinforcements	
Betoniraudoitteet. Teräsbetonirakenteissa käytettävät raudoitteet	Concrete reinforcement. Rebars used in steel concrete structures	SFS 1267
Jänneteräket	Tendon steels	SFS 1265-1, SFS 1265-3
90.012 Betonituotteet	90.012 Concrete products	
Valmisbetoni ja jännemenetelmät	Ready-mix concrete and tendon methods	EN 206, SFS 7022
Betonipintojen suojaus- ja korjausaineet	Protection and repair agents for concrete surfaces	EN 1504-2...6
Lattiatasoitteet	Floor screeds	EN 13813
90.013 Lämmöneristeet	90.013 Thermal insulation	
mineraalivillat	mineral wools	EN 13162
EPS-eristeet	EPS insulation	EN 13163
suulakepuristetut polystyreenituotteet (XPS)	extruded polystyrene products (XPS)	EN 13164
polyuretaanituotteet	polyurethane products	EN 13165
tehdasvalmisteiset fenoli- vaahtotuotteet (PF)	factorymade phenolic foam products (PF)	EN 13166

vahtolasituotteet (cellular glass)	foam glass products (cellular glass)	EN 13167
lastuvillatuotteet	chip wool products	EN 13168
puukuitutuotteet	wood fiber products	EN 13171
kevytsoratuotteet	Light gravel products	EN 14063
puhallusvillatuotteet	blown wool products	EN 14064
ruiskutettavat PUR-eristeet	sprayable PUR insulations	EN 14315-1
lämmöneriste- ja kevyttäytetuotteet maa- ja vesirakentamiseen	thermal insulation and light-weight filling products for land and water construction	
paisutetut polystyreenituotteet (EPS)	expanded polystyrene products (EPS)	EN 14933
suulakepuristetut polystyreenituotteet (XPS)	extruded polystyrene products (XPS)	EN 14934
kevytsora	Light gravel	EN 15732
90.014 Rakennuslasi	90.014 Building glass	
eristyslasit	laminated glasses	EN 14440
karkaistut lasit	insulating	EN 12150-2
laminoidut lasit	glasses - tempered glasses	EN 1279-5
90.015 Kalvot, kelmut, geotekstiilit, vesieristeet	90.015 Films, foils, geotextiles, waterproofing	
kattojen bitumiset vedeneristeet	bituminous roof waterproofing	EN 13707
muovi- ja kumikelmut kattojen vesieristykseen	plastic and rubber films for roof waterproofing	EN 13956
bitumiset kosteuseristeet	bituminous moisture insulation	EN 13969
bitumiset höyrynsulut	bituminous vapor barriers	EN 13970
muoviset ja kumiset höyrynsulut	plastic and rubber vapor barriers	EN 13984
polymeerimodifioidut bitumit	polymermodified bitumens	EN 14023
nestemäisenä levitettävät vedeneristeet laatoituksien alle	liquid-applied waterproofing under tiles	EN 14891
bitumiset kosteuseristekaistat	bituminous moisture insulation strips	EN 14967
geotekstiilit	geotextiles	EN 15381; EN 13251; EN 13252; EN 13253
90.016 Kipsituotteet	90.016 Gypsum products	
tavanomaiset kipsilevyt	conventional plasterboards	EN 520
kipsilevytuotteet alaslaskettuihin kattoihin	plasterboard products for suspended ceilings	EN 14246
mekaaniset kiinnikkeet kipsilevyihin	mechanical fasteners for plasterboards	EN 14566
metallirangat kipsilevyjärjestelmiin	metal rods for plasterboard systems	EN 14195
mattovahvisteiset kipsilevyt	carpet-reinforced plasterboards	EN 15283-1
kuituvahvisteiset kipsilevyt	fiber-reinforced plasterboards	EN 15283-2

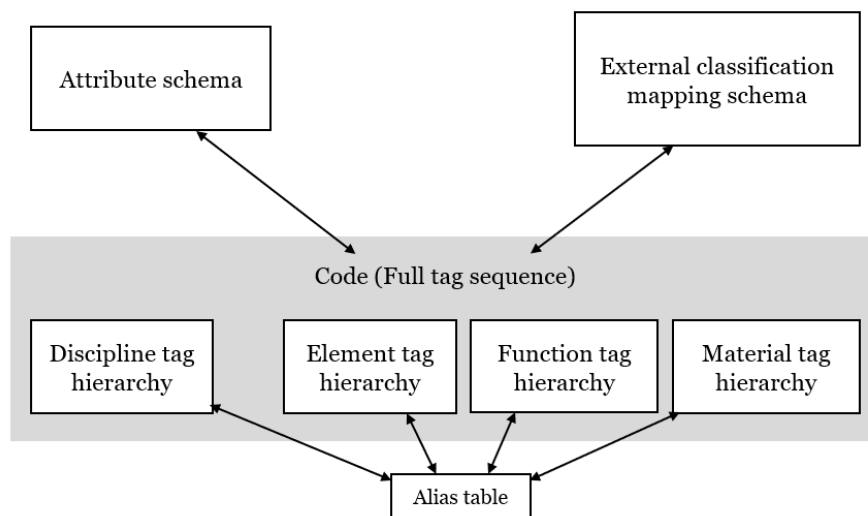
90. 021 Kantavat ja osittain kantavat rakennustuotteet	90. 021 Load-bearing and partially load-bearing construction products	
Betonielementit ja kevytbetonielementit	Concrete elements and lightweight concrete elements	
ontelolaatat	hollow tiles	EN 1168 SFS 7016
perustuspaalut	foundation piles	EN 12794
ripalaattaelementit	ribbed slab elements	EN 13224 SFS 7026
pilari- ja palkkielementit	column and beam elements	EN 13225 SFS 7026
kuorilaatat	shell tiles	EN 13747 SFS 7026
betoniporraselementit	concrete stair elements	EN 14843 SFS 7026
betoniseinäelementit	concrete wall elements	EN 14992 SFS 7026
harvasta kevytsorabetonista valmistetut kevytbetonielementit	manufactured lightweight concrete elements from light gravel	EN 1520
höyrykarkaistut kevytbetoni elementit	steam hardened lightweight concrete elements	EN 12602
Rakenteelliset teräs/alumiinituotteet ja tuotejärjestelmät	Structural steel/aluminum products and product systems	
teräs- ja alumiinirakenteet	steel and aluminum structures	EN 1090-1
kuumavalssatut rakenneteräkset	hotrolled structural steels	EN 10025-1
putkiprofiilit	pipe profiles	EN 10219-1
ruostumattomat profiilit	stainless profiles	EN 10088-5
Rakenteelliset puutuotteet	Structural wood products	
puurakenteet, liimapuu	wooden structures, glulam	EN 14080
naulalevyliitoksin kootut puuelementit	wooden elements assembled with nail plate joints	EN 14250
viilupuu LVL	LVL veneer	EN 14374
sormijatkettu sahatavara	fingerjointed sawn timber	EN 15497
puulevyt	wooden boards	EN 13986
Laakerit	Bearings	
kumilevyalaakerit	rubber plate bearings	EN 1337-3
rullalaakerit	roller bearings	EN 1337-4
kumipesälaakerit	rubber housing bearings	EN 1337-5
keinulaakerit,	rocker bearings	EN 1337-6
kalotti- ja sylinterilaakerit PTFE-liukupinnoilla	cap and cylinder bearings with PTFE sliding surfaces	EN 1337-7
ohjauslaakerit ja estolaakerit	Guide bearings and blocking bearings	EN 1337-8
Muuratut tuotteet	Masonry products	
muurauskappaleet	masonry pieces	EN 771-1...6 SFS 7001
muurauslaastit	masonry mortars	EN 998-2 SFS 7001
muuraussiteet	masonry ties	EN 845-1 SFS 7001

puolirakenteelliset aukon ylityspalkit	semi-structured opening crossing beams	EN 845-2 SFS 7001
90. 031 Katteet	90. 031 ???Margins ?Covers	
betonikattotiilet	concrete roof tiles	EN 490
kattokulkutiet, kattosillat, askel-tasot	roof walkways, roof bridges, steps	EN 516
itsekantavat metalliohutlevytuotteet vesikattoihin ja seinien verhouksiin	self-supporting sheet metal products for water ceilings and wall coverings	EN 14782
ei-kantavat metalliohutlevytuotteet vesikattoihin ja seinien verhouksiin	non-load-bearing sheet metal products for water ceilings and wall coverings	EN 14783
Aluskatteet	Underlays	
epäjatkuvien katteiden aluskatteet	underlays for discontinuous roofs	EN 13859-1
90.032 Seinien, lattioiden ja sisäkattojen pintarakenteet, julkisivujärjestelmät	90.032 Surface structures of walls, floors and interior ceilings, facade systems	
Sisäverhoukset, julkisivuverhoukset ja julkisivujärjestelmät	Interior cladding, facade cladding and facade systems	
julkisivujärjestelmät	facade systems	EN 13830
lattiapäällysteet	floor coverings	EN 14041
alakatot	false ceilings	EN 13964
puuverhoukset	wood cladding	EN 14915
90. 033 Metallisandwich-paneelit	90. 033 Metal sandwich panels	
itsekantavat metalliohutlevypinta-aiset sandwich-elementit	self-supporting sandwich elements with sheet metal surfaces	EN 14509 SFS 7030
90. 034 Puulevyt ja kuitusementtilevyt	90. 034 Wood boards and fiber cement boards	
puulevyt sisä- ja ulkokäyttöön	wooden boards for indoor and outdoor use	EN 13986
kuitusementtitasolevyt	fiber cement level boards	EN 12467
90.035 Ovet, ikkunat, lukot, helat	90.035 Doors, windows, locks, hinges	
ikkunat ja sisäänkäyntiovet	windows and entrance doors	EN 14351-1
90.051 Palosuojajärjestelmät	90.051 Fire protection systems	
savusulut	smoke barriers	EN 12101-1 SFS 7023
savunpoistoluukut	smoke exhaust hatches	EN 12101-2 SFS 7024
savunhallintakanavat	smoke control ducts	EN 12101-7 SFS 7028
90.062 Putket ja säiliöt muuhun kuin juomavesikäyttöön	90.062 Pipes and tanks for other than drinking water use	INSTA SBC PS 116 and NPG PS 116

Appendix 5. Classification system prototype – full hierarchies and tables with explanations

Overview and code explanation

The full classification system prototype includes: Element tag hierarchy, Functional tag hierarchy, Material tag hierarchy, Discipline tag hierarchy, Classification system mapping schema, Attribute schema, Translation alias table.



A conventional class is substituted by a full tag sequence that is represented by a code. The full tag sequence is comprised of any combination of tags, that should follow the hierarchy order. The attribute schema and the classification mapping schema link to a code snippet that is being compared against the full code (tag sequence). The alias table is only responsible for tag name translations.

The full tag sequence code is consisting of tag labels (e.g. FL for Floors). The labels form a code, in which each position is reserved for a label or their hierarchical sequence of a particular hierarchy.

[Discipline]-[Element hierarchy]-[Function hierarchy]-[Material hierarchy]

Each position can hold an unlimited number of label sequences or code snippets delimited by “_”, while separate position are separated by “-”.

[Discipline]-[Element hierarchy 1]_[Element hierarchy 2]..._[Element hierarchy N]-[Function hierarchy]-[Material hierarchy]

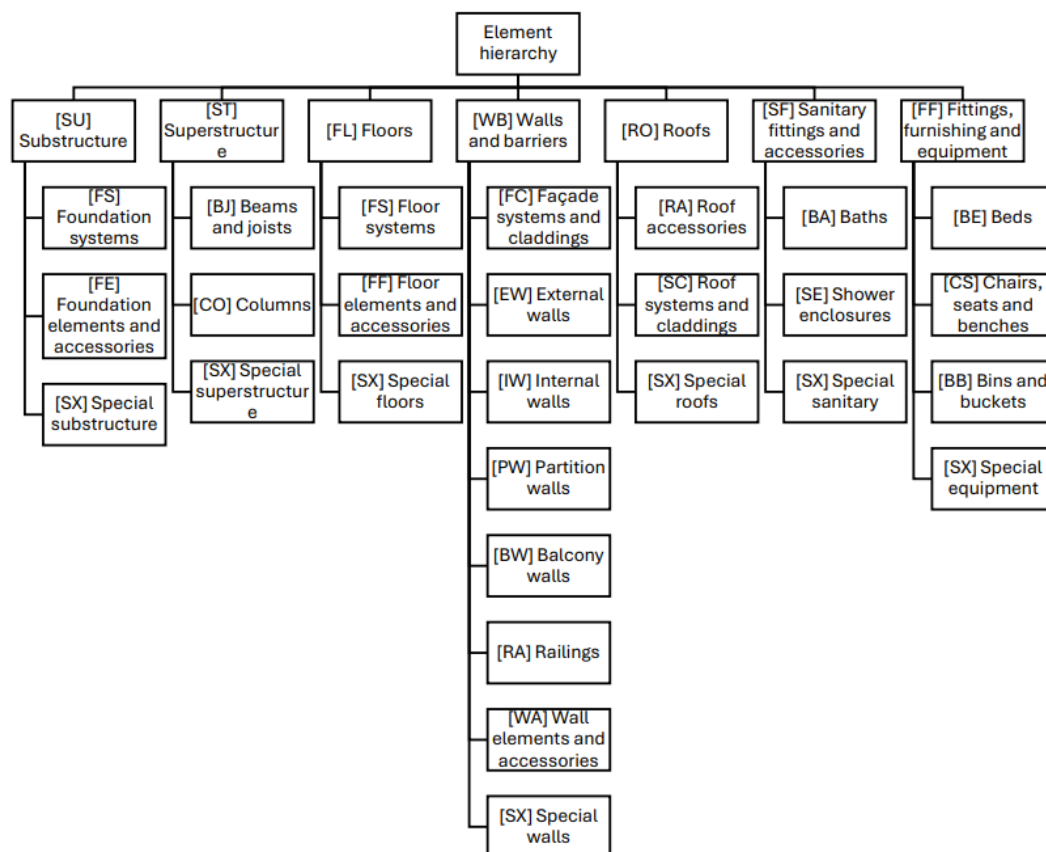
An unused position in the full code is marked with “XX”. Thus, an example of the code would look like:

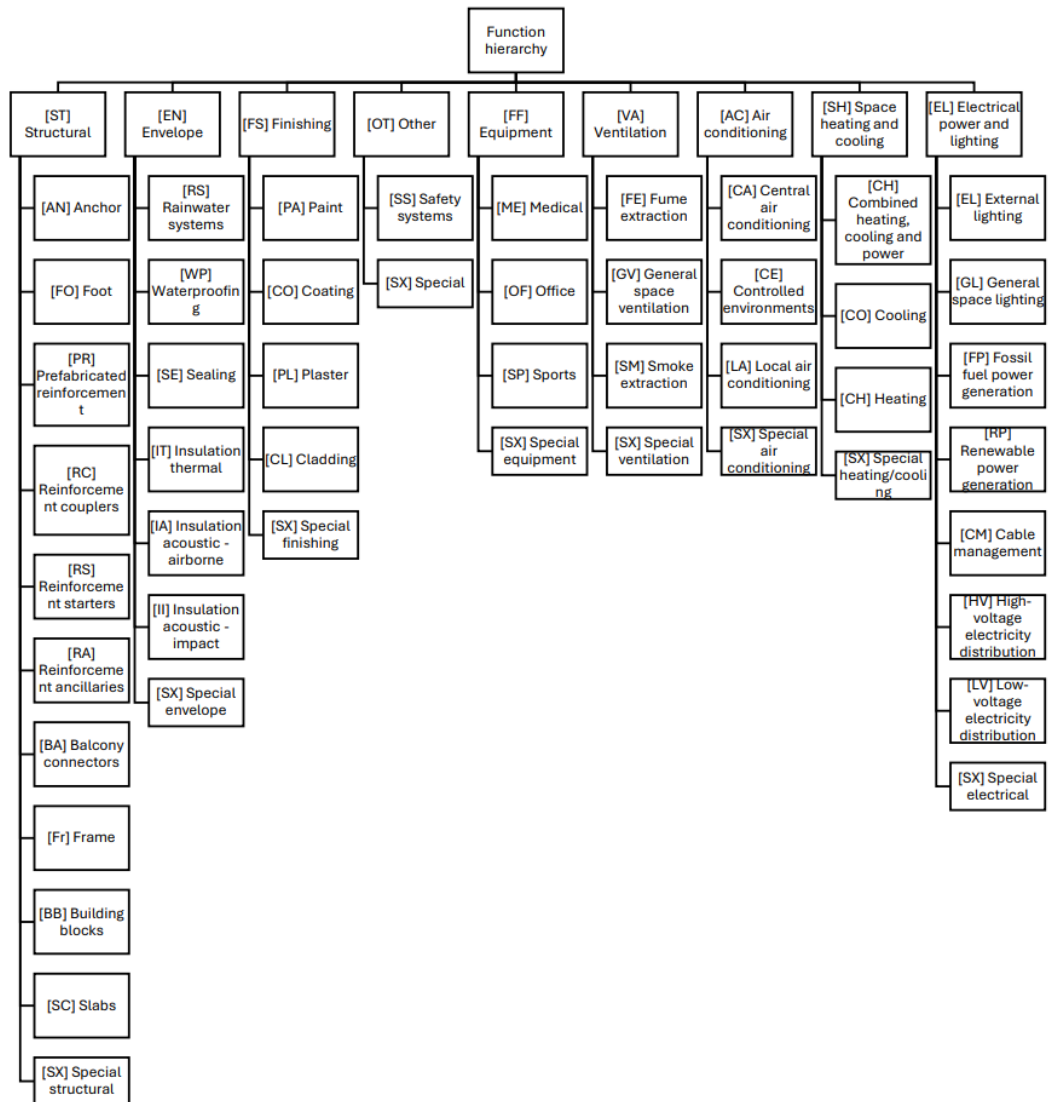
S-RORA-XX-MEALAS_MEALAP

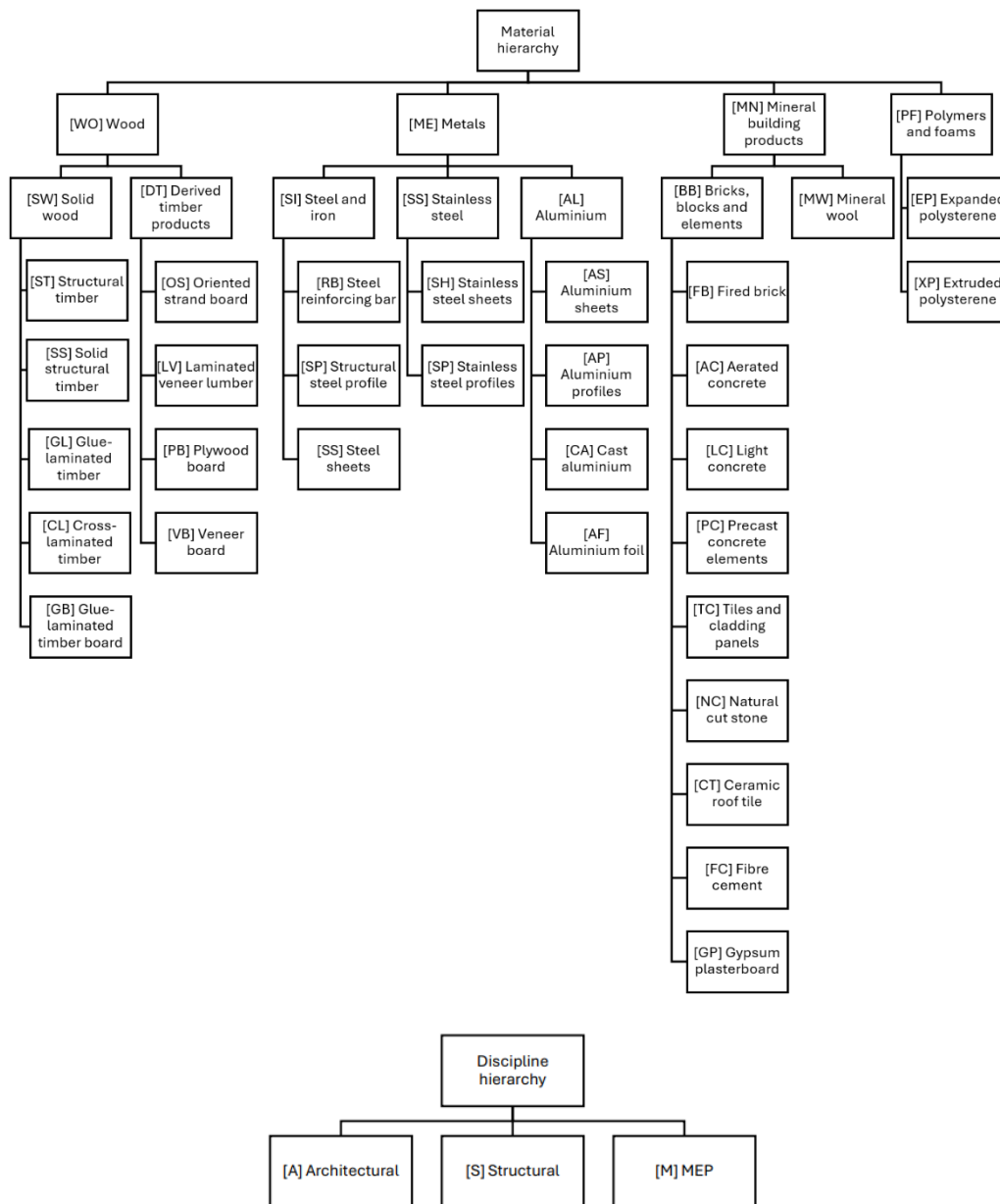
This code is representing the following tags:

- Structural discipline
- Roofs – Roof accessories
- No function tag
- Metals – Aluminium – Aluminium sheets
- Metals – Aluminium – Aluminium profiles

Taxonomy as tree structures







Taxonomy as lists

Element Hierarchy

SU – Substructure

FS – Foundation systems

FE – Foundation elements and accessories

SX – Special substructure

ST – Superstructure

BJ – Beams and joists

CO – Columns

SX – Special superstructure

FL – Floors

FS – Floor systems

FF – Floor elements and accessories

SX – Special floors

WB – Walls and barriers

FC – Façade systems and claddings

EW – External walls

IW – Internal walls

PW – Partition walls

BW – Balcony walls

RA – Railings

WA – Wall elements and accessories

SX – Special walls

RO – Roofs

RA – Roof accessories

SC – Roof systems and claddings

SX – Special roofs

SF – Sanitary fittings and accessories

BA – Baths

SE – Shower enclosures

SX – Special sanitary

FF – Fittings, furnishing and equipment

BE – Beds

CS – Chairs, seats and benches

BB – Bins and buckets

SX – Special equipment

Function Tags Hierarchy

ST – Structural

AN – Anchor

FO – Foot

PR – Prefabricated reinforcement

RC – Reinforcement couplers
RS – Reinforcement starters
RA – Reinforcement ancillaries
BA – Balcony connectors
Fr – Frame
BB – Building blocks
SC – Slabs
SX – Special structural

EN – Envelope

RS – Rainwater systems
WP – Waterproofing
SE – Sealing
IT – Insulation thermal
IA – Insulation acoustic – airborne
II – Insulation acoustic – impact
SX – Special envelope

FS – Finishing

PA – Paint
CO – Coating
PL – Plaster
CL – Cladding
SX – Special finishing

OT – Other

SS – Safety systems
SX – Special

FF – Equipment

ME – Medical
OF – Office
SP – Sports
SX – Special equipment

VA – Ventilation

FE – Fume extraction
GV – General space ventilation
SM – Smoke extraction

SX – Special ventilation

AC – Air conditioning

CA – Central air conditioning

CE – Controlled environments

LA – Local air conditioning

SX – Special air conditioning

SH – Space heating and cooling

CH – Combined heating, cooling and power

CO – Cooling

CH – Heating

SX – Special heating/cooling

EL – Electrical power and lighting

EL – External lighting

GL – General space lighting

FP – Fossil fuel power generation

RP – Renewable power generation

CM – Cable management

HV – High-voltage electricity distribution

LV – Low-voltage electricity distribution

SX – Special electrical

Material Tags

WO – Wood

SW – Solid wood

ST – Structural timber

SS – Solid structural timber

GL – Glue-laminated timber

CL – Cross-laminated timber

GB – Glue-laminated timber board

OS – Oriented strand board

LV – Laminated veneer lumber

PB – Plywood board

VB – Veneer board

ME – Metals

SI – Steel and iron
RB – Steel reinforcing bar
SP – Structural steel profile
SS – Steel sheets
SS – Stainless steel
SH – Stainless steel sheets
SP – Stainless steel profiles
AL – Aluminium
AS – Aluminium sheets
AP – Aluminium profiles
CA – Cast aluminium
AF – Aluminium foil

MN – Mineral building products

BB – Bricks, blocks and elements
FB – Fired brick
AC – Aerated concrete
LC – Light concrete
PC – Precast concrete elements
TC – Tiles and cladding panels
NC – Natural cut stone
CT – Ceramic roof tile
FC – Fibre cement
GP – Gypsum plasterboard

IN – Insulation materials

MW – Mineral wool
EP – Expanded polystyrene
XP – Extruded polystyrene

Discipline Tags

A – Architectural
S – Structural
M – MEP

Mapping tables and principle of mapping to other classification systems

Talo 2000

Talo 2000		Prototype	
Code	Name	Codes	Tag sequences
1212	Enclosure walls, foundation columns, foundation beams	SUFS	Substructure - Foundation systems
1211	Footings	SUFE	Substructure - Foundation elements and accessories
1213	Special foundations	SUSX	Substructure - Special substructure
1241	External walls	WBEW	Walls and barriers - external walls
		WBFC	Walls and barriers - Façade systems and claddings
1262	Roof substructures	ROSC, ENWP	Roofs - Roof systems and claddings, Envelope - Waterproofing
		ROSC, ENSE	Roofs - Roof systems and claddings, Envelope - Sealing
		ROSC, ENIT	Roofs - Roof systems and claddings, Envelope - Insulation thermal
		RORA	Roofs - Roof accessories
1263	Roofings	ROSC	Roofs - Roof systems and claddings

Uniclass

Uniclass		Prototype	
Code	Name	Codes	Tag sequences
EF_20_05_30	Foundations	SUFS	Substructure - Foundation systems
EF_25_10_25	External walls	WBEW	Walls and barriers - external walls
Pr_25_57_06_53	Mineral wool insulation	ENIT, INMW	Envelope - Insulation thermal, Insulation materials - Mineral wool
Pr_20_29_03_86	Structural anchors	STAN	Structural - Anchor

The prototype uses containment mapping rather than one-to-one code equality. A prototype full tag sequence represented by a code (e.g. AS-RORA-XX-MEALAS_MEALAP) is matched to an external class if it contains the required prototype code snippet specified by a mapping rule (e.g. RORA).

- A rule keyed by a prototype code snippet matches any code (full tag sequence) that contains the snippet, irrespective of other tags or order within the full tag sequence.
- A key separated by commas (e.g. ROSC, ENIT) expresses a conjunction and means that the target code must contain all listed code snippets to match.

- If an external class has multiple mapped code snippets, any of them should satisfy the match if present within the full tag sequence.
- A single prototype code snippet can be mapped to multiple external classes and vice versa. This relation is intentional and represents practical overlaps.

To resolve conflicts in situations where the external class is too broad for the prototype, matches with more duplicate code entries are preferred. With the presence of multiple deeper tags mapped the parent tag is considered the closest match.

External classes can be interlinked via shared prototype tags. If two external classification categories map to the same prototype code, they are considered cross-linked.

When no exact match is present either between an external class and the prototype or between two external classifications via the shared prototype code, the nearest match shall be proposed ranked by code overlap.

Attribute table and principles of creation and mapping

Attributes				
Name	Unit	Values range	Code	Tag sequence
Width	m		XX	Any
Length	m		XX	Any
Global Warming Potential (GWP100), A1-A3, EPD Total	kg CO ₂ -eq		XX	Any
Reaction to fire		A1...F	XX	Any
Wind load resistance	kN/m ²		WBFC	Walls and barriers - Façade systems and claddings
Thermal resistance	m ² ·K/W		ENIT, WBFC, WBEW, WBBW, ROSC, FLFS	Envelope - Insulation thermal, Walls and barriers - Façade systems and claddings, Walls and barriers - External walls, Walls and barriers - Balcony walls, Roofs - Roof systems and claddings, Floors - Floor systems

The Attributes schema defines which properties apply to which items in the prototype. Some attributes are generic and reflect the geometry or parameters of the item (Width, Length). Sustainability metrics are global and universal attributes (GWP in the table). Parameters such as “Reaction to fire” and “Thermal resistance” are in accordance with the European standard naming and units (or format). These parameters were taken from respective standards for the applicable codes.

An attribute applies to an object if the object's prototype tag sequence contains at least one of the code snippets listed in Code (containment). If an attribute must apply only when multiple tags co-occur, define a separate rule using those tags together (logical AND). Matching is order- and axis-agnostic (it does not matter whether a tag comes from the element, function, or material hierarchy). Because attributes are attached to prototype tags, they automatically transfer with the object when translated to Talo 2000, Uniclass, or other systems via the mapping rules.

Note

Translation tables have been left out of this paper, however translations based on commonly used and understood locally terminology are part of the concept. The translations can be implemented via a simple alias table, where values are based on local classification system naming or standards.